Knowledge-Aided Space-Time Adaptive Processing for Functional Ultrasound: Robust Microvascular Detection in High-Clutter Regimes

Arthur Shune*
Independent Researcher

Abstract

Functional ultrasound (fUS) and ultrafast Doppler imaging face severe interference from tissue motion, phase aberrations, and aliased high-velocity flow, which can obscure slow microvascular dynamics. Low-rank clutter suppression methods such as motion-compensated singular value decomposition (MC–SVD) are widely used, but can degrade in extreme clutter and in the far right tail of flow evidence.

Motivated by airborne radar and RF array processing, we introduce a reduced-dimensional, post-beamformed space–time adaptive processing (STAP) framework for fUS slow-time flow detection. The method operates on localized spatiotemporal tiles with a fixed-parameter operating profile (tile geometry, slow-time aperture, Doppler band geometry, and covariance conditioning) and produces a matched-subspace STAP score, applied to the residual after a coarse baseline clutter filter. An optional knowledge-aided (KA) spatial-spectral prior uses local band-energy features to apply a conservative, shrink-only penalty in high-risk regions while protecting a high-confidence set; by construction, the post-KA score is non-increasing on penalized pixels. We also study conditional execution as a decision-rule variant that skips STAP outside a flow-proxy region and quantify its leakage sensitivity.

Quantitative detection performance is evaluated using a strict low-FPR operating discipline (typically 10^{-4}–10^{-3}, subject to empirical support), reflecting the sensitivity of downstream functional mapping to rare flow artifacts and consistent with tail-rate control philosophies in radar detection and tracking. We report results on labeled k-Wave brain simulations and on structurally labeled *in vitro* phantom acquisitions (Gammex; labels derived from B-mode geometry), and we benchmark against MC–SVD, adaptive/global SVD clutter filters (including spatial-singular similarity cutoffs), block-wise local SVD clutter filters, robust PCA, and tensor HOSVD baselines as well as conventional short-ensemble Doppler scores. We further report labeled within-ensemble motion ladders on phantom data via per-shot motion injection, and label-free stability under controlled motion corruption on kilohertz ultrasound localization microscopy (ULM) IQ blocks. Because most open in-vivo fUS releases do not preserve raw high-PRF complex IQ with independent H_1/H_0 labels, we restrict quantitative ROC claims to labeled simulations and structurally labeled phantoms, using in-vivo data only for robustness and telemetry audits. All experiments and figures are accompanied by a command-level reproducibility manifest in the Supplementary Material (exact scripts, arguments, and expected artifact paths).

*Work performed in personal capacity. Correspondence: `arthur.shune@gmail.com` (alt: `arthur@skymesasystems.com`). ORCID: `0009-0004-6433-3856`. Current employment: BlackSky, Inc. Founder: SkyMesa Systems; Valence Intelligence Systems. No external funding.

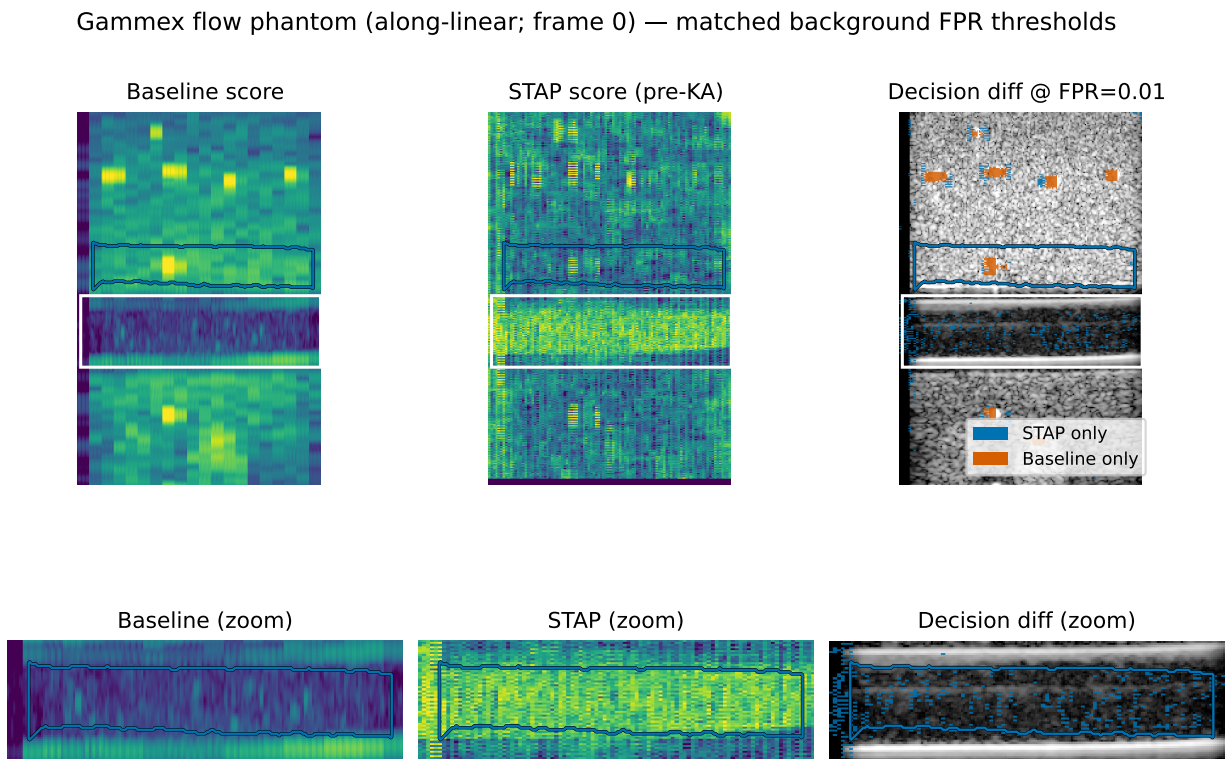

Figure 1: Leading structural-fidelity visual on a structurally labeled Gammex flow phantom acquisition (labels derived from B-mode-only tube segmentation). Top row: baseline score map S_{base} (SVD band-pass clutter suppression followed by power Doppler), STAP pre-KA matched-subspace score $S_{\text{stap,pre}}$, and a matched-FPR decision-difference panel at a fixed background false-positive rate (10^{-2}; chosen here for visualization), showing pixels detected by STAP only versus baseline only. Bottom row: zoomed region of interest. The tube contour is overlaid for reference.

1 Introduction

Microvascular flow detection in ultrasound is an extreme-tail problem: flow evidence occupies a small fraction of pixels while tissue motion, phase aberration, and aliased high-velocity clutter induce heavy-tailed background statistics. In this regime, improvements in global summaries such as AUC can be operationally uninformative; the key requirement is stable detection at ultra-low false-positive rates.

We address this setting with a localized, STAP-inspired detector in which adaptation is temporal (slow-time), while spatial localization enters through tile-local covariance estimation. Instead of relying only on global low-rank suppression, we whiten local slow-time snapshots and score energy in a flow subspace, targeting spatially varying interference patterns that dominate extreme-tail false alarms. Figure 1 provides a motivating qualitative example on a structurally labeled flow phantom: at a matched background-FPR operating point, STAP modifies detection decisions in a way that is consistent with the B-mode-derived vessel geometry.

Optionally, we add a conservative shrink-only regularization layer (knowledge aid) that compresses extreme-tail artifacts without score inflation: post-regularization scores are pointwise non-

increasing, and an explicit do-not-penalize protected set is preserved (details in the Supplementary Material).

Contributions.

- A reduced-dimensional, post-beamformed temporal STAP-like detector: tile-local covariance adaptation in slow time with a matched-subspace flow-band score, evaluated at strict low-FPR operating points.
- A strict-tail evaluation discipline with empirically supported FPR floors and baseline comparators including adaptive/global and local eigen/SVD clutter filters.
- A leakage-controlled conditional-execution variant treated explicitly as a changed decision rule (not a cost-free speedup), with reproducible GPU latency reporting.
- An optional shrink-only knowledge-aided (KA) tail-regularization layer with a protected set and a monotonicity guarantee (details in the Supplementary Material).

Scope and claims. Claims are stratified by data regime and summarized in Table 2. Quantitative detection efficacy is limited to regimes with independent H_1/H_0 labels; unlabeled in-vivo IQ and PD-only datasets are used for instrumentation and robustness analyses.

Organization. Section 2 defines the detection problem and score taxonomy. Section 3 presents the localized detector and summarizes the optional shrink-only KA regularization (full KA machinery in the Supplementary Material). Section 4 specifies fixed-profile experimental design and integrity controls. Section 5 reports results, followed by discussion and conclusion.

2 Background and Problem Formulation

Problem setup. For each spatial location (or tile-aggregated snapshot), let $x \in \mathbb{C}^{L_t}$ denote a slow-time vector. We consider

$$H_0 : \; x = c + n, \qquad H_1 : \; x = s + c + n,$$

where s is flow-consistent signal, c is structured clutter/motion, and n is residual noise. In practice, H_0 is spatially nonstationary and heavy-tailed, motivating local covariance adaptation and tail-focused evaluation.

Core notation and bands. We use band operators (P_o, P_f, P_g, P_a) for clutter/DC, flow, guard, and alias regions, respectively. For short ensembles, these are exact orthogonal DFT-bin projectors; for longer apertures they are leakage-controlled analysis operators and are treated as band-energy statistics with explicit sanity sentinels (Supplementary Material).

2.1 Related context

STAP and knowledge-aided STAP. Classical MVDR/LCMV/STAP and knowledge-aided variants shape covariances using external structure [47–50, 55, 56, 59]. In radar, KA is often expressed as covariance shaping/blending or knowledge-aided covariance estimation; here we adopt the KA spirit but implement it as a shrink-only, fail-safe score regularizer on top of a fixed detector statistic, with explicit conditions for inert behavior out of regime.

Ultrasound clutter suppression. Spatiotemporal SVD clutter filters are a standard tool in ultrafast Doppler and fUS [6, 7, 11, 12, 17]. Real-time feasibility has been studied via randomized/approximate decompositions and downsampling strategies [16], as well as GPU-parallel eigen-based clutter filtering frameworks [15]. Robust PCA, unfolded/learned RPCA, and tensor/HOSVD variants have also been studied for clutter suppression and motion robustness [23, 24, 41]. More generally, when raw channel/aperture data are available, aperture-domain multidimensional clutter filtering can provide additional degrees of freedom beyond beamformed slow-time processing [40]. We benchmark STAP against MC–SVD and established adaptive/local SVD clutter filters (adaptive cutoffs via spatial singular-vector similarity and block-wise local SVD), as well as RPCA and HOSVD baselines; conventional short-ensemble Doppler comparators are additionally reported on phantom data. These baselines are included to represent state-of-practice eigen/SVD clutter filtering beyond fixed-cutoff global Casorati SVD and to preempt the common critique that strict-tail gains are attributable only to localization or cutoff tuning rather than to the detector formulation and tail operating discipline.

Relation to eigen-based and local adaptive clutter filtering. Eigen-based clutter filter design for ultrasound color flow imaging and block-wise/local adaptive clutter filters are closely related ultrasound-native approaches to covariance/eigenspace adaptation [13–15]. Song *et al.* demonstrate local clutter filtering in *in vivo* settings (e.g., kidney imaging), while many open fUS releases do not preserve raw high-PRF complex IQ suitable for labeled Doppler-side detector evaluation. We therefore use labeled k-Wave pilots and structurally labeled phantoms for quantitative ROC claims and use *in vivo* data for robustness and instrumentation audits. Our contribution differs in the *detector* formulation and evaluation discipline: in contrast to approaches framed primarily as clutter suppression followed by power Doppler scoring, we compute a GLRT-motivated matched-subspace detector score on tile-local slow-time snapshots with explicit Doppler band geometry (flow/guard/alias) and optional tile-local whitening, evaluate at empirically supported ultra-low FPR operating points where background-tail behavior dominates, and treat compute gating (conditional execution) as a distinct decision rule with leakage controls.

ULM pipelines and motion correction. Ultrafast ultrasound localization microscopy (ULM) achieves deep super-resolution vascular imaging by localizing sparse microbubble echoes over high-PRF ensembles [37]. Robust clutter suppression and motion correction are central to ULM pipelines, including subwavelength motion correction tailored to ultrafast acquisitions [38]. We do not claim improved microbubble localization performance here; rather, we use a public kHz IQ ULM benchmark dataset to report label-free stability and motion robustness of Doppler-side score maps under controlled motion corruption, illustrating how a tail-stable detector can complement upstream motion correction and downstream localization stages.

Twinkling artifact and Doppler phantoms. The Doppler "twinkling artifact" has been documented as a persistent color-flow phenomenon on calcifications and other strong reflectors, and is widely used as a diagnostic benchmark for Doppler processing chains. Leonov *et al.* provide an open-access database of raw beamformer signals with phantoms including a flow-through Gammex phantom and calculi/metal/plastic objects [39]. We use these data to (i) obtain a structurally labeled phantom ROC via B-mode-only tube segmentation and (ii) construct an artifact-heavy KA hygiene regime where the prior penalty is expected to be conservative and auditable.

fUS detection and hemodynamic analysis. GLM-based detection and atlas-driven ROI analysis are common in hemodynamic fUS [42,43]; our hemodynamic STAP variant reuses the same banded temporal machinery on PD time series and adds an alias-aware shrink-only prior penalty

Symbol / name	What it measures	Used for	Data requirement
S_{base}	baseline ROC score map on X_{base} (PD power; higher = more flow-like)	baseline ROC / low-FPR tables	IQ
$S_{\text{stap,pre}}$	pre-KA matched-subspace ratio STAP score map	STAP ROC / low-FPR tables	IQ + local covariance
$S_{\text{stap,post}}$	post-KA shrink-only score map $wS_{\text{stap,pre}}$	KA behavior audits	IQ + feature map
$\text{PD}^{\uparrow}$ (display PD), masks	PD display/proxy map and derived masks	visualization and deterministic preprocessing (gating/protected sets)	IQ (or PD-only proxy mode)

Table 1: Score/map taxonomy used throughout. ROC score maps are kept separate from display/proxy maps used only for visualization and deterministic preprocessing. Display/proxy maps are never used to compute ROC/TPR/FPR (only to define gating/protected-set masks, which are audited separately).

as a safety layer.

2.2 Baseline clutter suppression and score conventions

Beamformed slow-time data. Let $X[t, y, x] \in \mathbb{C}$ denote a beamformed slow-time IQ movie over $t = 0, \ldots, T-1$ and spatial indices (y, x). We first apply a baseline clutter-suppression operator (e.g. motion-compensated SVD), producing a residual cube $X_{\text{base}}[t, y, x]$ used for both baseline scoring and as the input to STAP.

Power Doppler proxy and baseline scores. From the baseline-filtered cube we define a conventional power Doppler (PD) brightness map used only as a display/proxy map for deterministic preprocessing (e.g. proxy masks) and visualization,

$$\text{PD}^{\uparrow}(y, x) = \frac{1}{T} \sum_{t=0}^{T-1} |X_{\text{base}}(t, y, x)|^2 .$$

For quantitative evaluation and thresholding we use *flow-evidence* score maps where higher values indicate stronger evidence of flow, and we calibrate thresholds on the right tail of these score distributions. We denote the baseline ROC score map as $S_{\text{base}}(y, x)$ (power Doppler on X_{base}). For short-ensemble regimes we additionally report auxiliary, conventional baseline comparators (e.g. log-power Doppler and Kasai lag-1 magnitude) to avoid baseline-score idiosyncrasies dominating conclusions (Notation). Note: $\text{PD}^{\uparrow}$ and S_{base} use the same underlying power statistic (up to fixed scaling); we keep separate names to make it explicit when PD is used only as a display/proxy map (masks/visuals) versus as a ROC score map.

2.3 Score taxonomy

Unless explicitly stated, "STAP" refers to the pre-KA matched-subspace score $S_{\text{stap,pre}}$; KA is analyzed as an optional, shrink-only post-processing regularizer.

3 Method

Algorithm 1 (STAP pipeline with optional KA regularization, fixed profile).

1. Baseline clutter suppression (e.g. MC–SVD) produces a residual cube X_{base} and baseline score $S_{\text{base}}(y, x)$.
2. For each tile Ω_i: form slow-time snapshots $\{x_{i,k}\}_{k=1}^{K}$ via spatial sampling and, optionally, a sliding-window temporal embedding (Hankel matrix construction), estimate conditioned covariance R_i, and whiten $y_{i,k} = R_i^{-1/2} x_{i,k}$.
3. Compute a matched-subspace ratio score (ACE/MSD-style) on whitened snapshots, e.g. $S_{\text{stap,pre}}(i) = \frac{1}{K}\sum_k \frac{y_{i,k}^{\text{H}} P_f\, y_{i,k}}{y_{i,k}^{\text{H}}(I-P_f)\, y_{i,k} + \rho\, y_{i,k}^{\text{H}} y_{i,k} + \epsilon}$, and aggregate over overlapping tiles to obtain $S_{\text{stap,pre}}(y, x)$.
4. If KA enabled: compute a deterministic, shrink-only weight $w(i) \in (0, 1]$ from auxiliary band-energy statistics and proxy-support features (details in the Supplementary Material), preserve a protected do-not-penalize set (scores unchanged), and output $S_{\text{stap,post}}(y, x) = w\, S_{\text{stap,pre}}(y, x)$. Else, $S_{\text{stap,post}}(y, x) = S_{\text{stap,pre}}(y, x)$.

In practice STAP is intended as a second-stage residual covariance whitener: a coarse baseline clutter filter removes gross stationary tissue clutter, and STAP targets the remaining structured, spatially varying interference that dominates extreme-tail false alarms. A STAP-only ablation on phantom data is provided in Supplementary Material.

3.1 Temporal STAP core and Doppler band geometry

Tile-based STAP. To mitigate the degraded tail sensitivity of global low-rank filters in high-clutter regimes, we apply a localized temporal STAP core on overlapping spatial tiles. We use "STAP" in a STAP-inspired sense: the adaptive filter operates in slow time, while spatial localization enters through tile-local covariance adaptation. Let Ω_i denote tile i and L_t the slow-time aperture. From X_{base} we construct a set of slow-time training snapshots $\{x_{i,k} \in \mathbb{C}^{L_t}\}_{k=1}^{K}$ for each tile via spatial sampling and, when applicable, a sliding-window temporal embedding (Hankel construction) in slow time. Here K denotes the number of training snapshots used to estimate the tile covariance; in the fully Hankelized case, each pixel contributes $N = T - L_t + 1$ snapshots and pooling across Ω_i yields $K = N|\Omega_i|$ (or a subsampled/capped variant). We estimate a local covariance

$$\widehat{R}_i := \frac{1}{K}\sum_{k=1}^{K} x_{i,k} x_{i,k}^{\text{H}},$$

apply conditioning/diagonal loading to obtain R_i, and whiten snapshots as $y_{i,k} := R_i^{-1/2} x_{i,k}$.

Training support and Hankel dependence. In the fully Hankelized case, each pixel in Ω_i contributes $N = T - L_t + 1$ overlapping slow-time snapshots; pooling across the $|\Omega_i|$ pixels yields $K = N|\Omega_i|$ tile-local training vectors. These Hankelized snapshots are temporally correlated due to overlap, so K should not be interpreted as an independent-sample count. Our robust covariance estimators do not require independence, but the effective support can be significantly smaller than K; we therefore optionally subsample Hankel columns (stride) and cap the number of columns used for training to bound computation and avoid overweighting highly correlated samples (fixed per regime; Supplementary Material).

Robust covariance estimation and conditioning. We whiten with a regularized covariance R_i rather than the raw SCM $\widehat{R}_i$. In the primary profile we use a Tyler-type M-estimator on the pooled tile snapshots, iterating

$$R_i^{(n+1)} \leftarrow \frac{L_t}{K} \sum_{k=1}^{K} \frac{x_{i,k} x_{i,k}^{\mathrm{H}}}{x_{i,k}^{\mathrm{H}} (R_i^{(n)})^{-1} x_{i,k}}, \qquad R_i^{(n+1)} \leftarrow \frac{L_t}{\mathrm{tr}(R_i^{(n+1)})} R_i^{(n+1)},$$

with at most 25 iterations and early stopping when $\|R^{(n+1)} - R^{(n)}\|_F / \|R^{(n)}\|_F < 10^{-4}$ (latency-only experiments may cap iterations; Supplementary Material). We then apply deterministic conditioning (shrinkage toward μI and diagonal loading $\lambda \mu I$, where $\mu = \mathrm{tr}(R)/L_t$ and λ is fixed per regime) to enforce target condition numbers. In very short-ensemble regimes we may fall back to a Huber IRLS covariance estimate for numerical stability (Supplementary Material).

Terminology: "space–time" in fUS STAP. In classical radar STAP, a snapshot stacks spatial array channels and temporal pulse taps, and adaptation uses training snapshots from nearby range bins [47, 59]. In beamformed fUS IQ we operate on a single complex slow-time sample per pixel, so we do not exploit array-manifold degrees of freedom; our space–time vector is the slow-time stack $x \in \mathbb{C}^{L_t}$, while "space" enters through *localized adaptation*: we estimate R_i from snapshots drawn from a small spatial neighborhood (tile) assumed approximately stationary over that region. This yields a reduced-dimensional, post-Doppler STAP-like detector: we whiten by $R_i^{-1/2}$ and score energy in an explicitly defined Doppler flow subspace using a matched-subspace statistic (Section 3), rather than framing the method solely as clutter suppression followed by power Doppler. We use the term "STAP" to emphasize these defining elements (local covariance adaptation, whitening, and subspace detection) while being explicit that channel-domain STAP/beamforming is out of scope for the present beamformed pipeline.

Figure 2 illustrates the local tile extraction and optional sliding-window temporal embedding (Hankel construction) used to form $\widehat{R}_i$ and contrasts it with the global Casorati stacking used by MC–SVD.

Band operators and matched-subspace STAP score. We partition the Doppler spectrum using band projectors (P_o, P_f, P_g, P_a) (Notation). For short ensembles, these are exact, orthogonal DFT-bin projectors with $P_o + P_f + P_g + P_a = I$; for longer apertures they are leakage-controlled operators that are approximately orthogonal and are treated as band-energy statistics. We define a matched-subspace STAP test statistic by flow-subspace energy on whitened snapshots, e.g.,

$$S_{\mathrm{stap,pre}}(i) := \frac{1}{K} \sum_{k=1}^{K} \frac{y_{i,k}^{\mathrm{H}} P_f \, y_{i,k}}{y_{i,k}^{\mathrm{H}} (I - P_f) \, y_{i,k} + \rho \, y_{i,k}^{\mathrm{H}} y_{i,k} + \epsilon},$$

and form a pixel-level score map $S_{\mathrm{stap,pre}}(y, x)$ by overlap-add aggregation of the contributing tile statistics. This pre-KA score is the primary STAP detector output used for quantitative reporting unless explicitly stated otherwise.

Regularization and self-normalization avoidance. If we whitened using the exact sample covariance formed from the same snapshots being scored, the statistic collapses: the whitened sample covariance becomes (approximately) identity and averaged subspace energies concentrate near

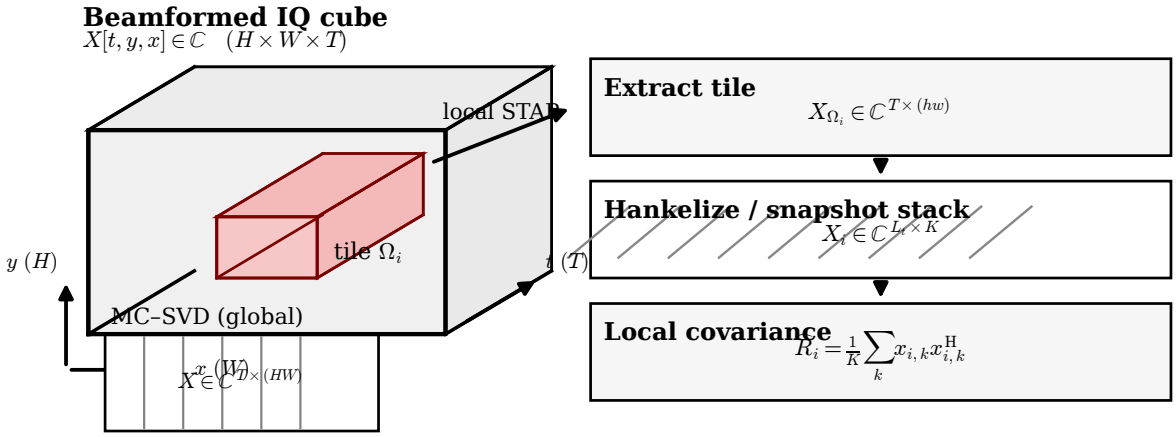

Figure 2: Schematic of sliding-window temporal embedding (Hankel matrix construction) and local covariance estimation. A localized spatial tile Ω_i is extracted from the beamformed slow-time data cube $X[t, y, x] \in \mathbb{C}$ $(H \times W \times T)$. Overlapping slow-time snapshots of length L_t are stacked (optionally Hankelized) to estimate the localized covariance matrix $\widehat{R}_i$. Unlike global low-rank filters (e.g. MC–SVD) that operate on a global Casorati matrix, this localized formulation preserves spatial-spectral heterogeneity of microvascular flow and highly structured tissue clutter.

constants (e.g. $\mathrm{tr}(P_f)$), destroying discrimination. We avoid this by whitening with a *regularized* covariance R_i (robust estimation plus shrinkage/diagonal loading), so that typically $R_i \neq \widehat{R}_i$ and the matched-subspace energy remains informative. We do not rely on an analytic null distribution for CFAR operation; operating points are empirically calibrated on labeled H_0 sets under the fixed-profile protocol of Section 4.1.

Matched-subspace statistic as a whitened subspace detector. Fix tile i and consider a single slow-time snapshot $x \in \mathbb{C}^{L_t}$. Under a Gaussian clutter-plus-noise model,

$$H_0 : \ x \sim \mathcal{CN}(0, R_i), \qquad H_1 : \ x = A_f s + n, \quad s \sim \mathcal{CN}(0, \sigma_s^2 I), \quad n \sim \mathcal{CN}(0, R_i),$$

where the columns of A_f span the flow subspace (equivalently the range of P_f). After whitening $y = R_i^{-1/2} x$, the GLRT for $\sigma_s^2 > 0$ versus $\sigma_s^2 = 0$ is monotone with the whitened subspace energy

$$T(x) = x^{\mathrm{H}} R_i^{-1/2} P_f R_i^{-1/2} x = y^{\mathrm{H}} P_f \, y,$$

which is a standard matched-subspace / adaptive energy detector. [57, 58] In implementation we report a self-normalized ACE/MSD-style ratio score that normalizes flow-subspace energy by the orthogonal complement (and a small ridge), which is monotone in $T(x)$ and serves as our pre-KA detector score $S_{\mathrm{stap,pre}}(i)$.

3.2 Optional shrink-only tail regularization (KA safety layer)

While the matched-subspace statistic $S_{\mathrm{stap,pre}}$ effectively isolates flow under standard colored-interference models, structured artifacts can still dominate extreme-tail operating points. We

therefore study an *optional* knowledge-aided (KA) layer that applies a conservative, shrink-only score penalty in high-risk regions. The KA mapping is deterministic and uses auxiliary band-energy and proxy-support statistics only as *side information*; proxies are never used to compute ROC/TPR/FPR (only to define protected/candidate sets and compute masks, which are audited separately).

KA in one picture (at-a-glance summary).

- **When it activates:** Only when deterministic sanity/support checks pass (valid band discretization; non-flat risk features; sufficient proxy support). Otherwise it is inert ($w \equiv 1$).
- **What it changes:** Applies a bounded, shrink-only weight $w(i) \in (0, 1]$ on an explicitly defined candidate set while preserving a protected do-not-penalize set (scores unchanged); the post-KA score is $S_{\text{stap,post}} = w\, S_{\text{stap,pre}}$.
- **What it cannot do:** It cannot increase any pixel score (no amplification) and proxies are not used for ROC/TPR/FPR reporting; any apparent operating-point uplift reflects background-tail compression under shrink-only mapping.
- **How it fails safely:** If out of regime, it defaults to "KA off" ($w \equiv 1$) and logs an explicit reason label, reducing to the pre-KA STAP detector.

Shrink-only mapping (guarantee). We implement knowledge aid via a bounded weight $w(i) \in (0, 1]$ applied in score space:

$$S_{\text{stap,post}}(i) = w(i)\, S_{\text{stap,pre}}(i), \qquad w(i) \in (0, 1].$$

Equivalently, define the per-tile scale $\gamma(i) := 1/w(i) \geq 1$ and write $S_{\text{stap,post}}(i) = S_{\text{stap,pre}}(i)/\gamma(i)$. The construction of $w(i)$ (feature definitions, proxy sets, regimes, and deterministic sentinels) is provided in Supplementary Material and Supplementary Material.

Lemma 1 (Shrink-only monotonicity)**.** *Let $w(i) \in (0, 1]$ and define $S_{\text{stap,post}}(i) = w(i)\, S_{\text{stap,pre}}(i)$. For any threshold τ,*

$$\{S_{\text{stap,post}} \geq \tau\} \subseteq \{S_{\text{stap,pre}} \geq \tau\}.$$

Consequently, on any subset A (e.g. H_0), the tail probability cannot increase at fixed τ:

$$\mathbb{P}\{S_{\text{stap,post}} \geq \tau \mid A\} \leq \mathbb{P}\{S_{\text{stap,pre}} \geq \tau \mid A\}.$$

Proof. If $S_{\text{stap,post}}(i) \geq \tau$, then $S_{\text{stap,pre}}(i) = S_{\text{stap,post}}(i)/w(i) \geq \tau$ since $w(i) \leq 1$. □

When reporting strict low-FPR operating points where τ is reselected post-KA to satisfy $\mathbb{P}\{S_{\text{stap,post}} \geq \tau \mid H_0\} = \alpha$, changes in TPR reflect background-tail compression despite scores never increasing.

3.3 Algorithmic integrity and deterministic safeguards

To make the architecture auditable and to avoid leakage or discretization artifacts masquerading as performance gains, we enforce deterministic, label-free safeguards:

- **Conditional execution as a decision-rule variant.** To reduce latency, STAP can be skipped outside a flow-proxy region and replaced by the baseline score there. Because this changes the overall decision rule, any conditional-execution reporting is treated as a distinct algorithmic variant and is accompanied by leakage controls (same-window vs. disjoint-window vs. randomized proxy masks). Unless otherwise stated, conditional execution uses *disjoint-window* proxy gating for detection reporting.
- **Short-ensemble band discretization sanity.** For extremely short ensembles (e.g. $N \in \{5, 9, 17\}$), the coarse frequency resolution $\Delta f = \mathrm{PRF}/N$ can cause requested bands to collapse or overlap after discretization. If the discretized bands are empty, DC-only, or overlapping, KA is disabled ($w \equiv 1$) with an explicit reason label.
- **Proxy-mask circularity control.** Proxy maps are never used to compute ROC/TPR/FPR; they are used only to define compute masks (conditional execution) or protected/candidate sets (KA), which are audited separately. Quantitative ROC claims are restricted to regimes with independent H_1/H_0 labels (simulator truth or B-mode structural geometry).

For auditability, we log explicit reason labels so that "KA off" ($w \equiv 1$) is an expected success mode under flat or non-actionable evidence. Supplementary Material provides modality-agnostic sufficient conditions and additional sentinels used to interpret when the KA prior should be active or inert.

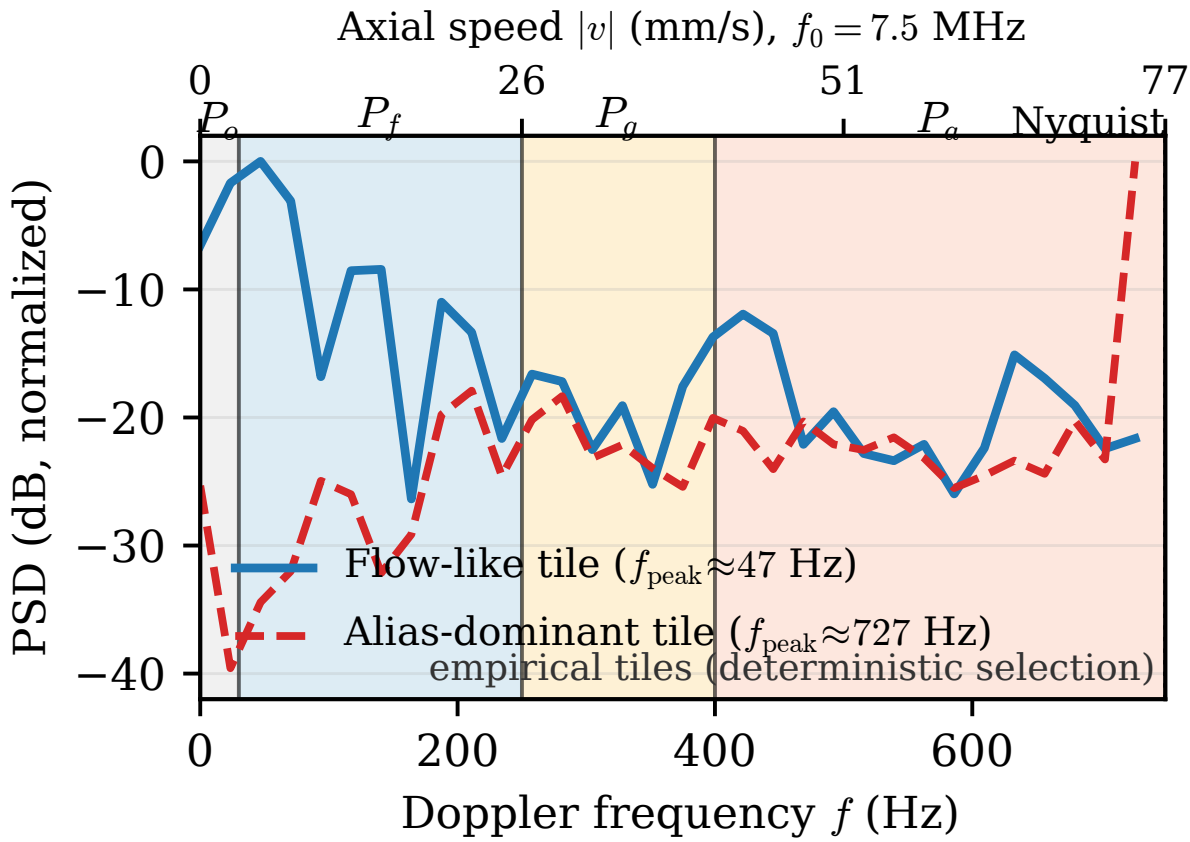

Figure 3: Doppler power spectral density (PSD) partitioning and kinematic band geometry (empirical example). Curves show one-sided PSDs of the tile-mean slow-time signal on a Brain-AliasContract k-Wave pilot window (PRF = 1500 Hz): a flow-like tile (peak in P_f) and an alias-dominant tile (peak in P_a). The Nyquist interval is deterministically partitioned into DC/high-amplitude clutter (P_o), microvascular flow (P_f), a transition guard band (P_g), and aliased high-velocity artifacts or vibration (P_a). Auxiliary band-energy descriptors (e.g. alias-to-flow ratios) are used for interpretation and, optionally, for shrink-only tail regularization (Supplementary Material).

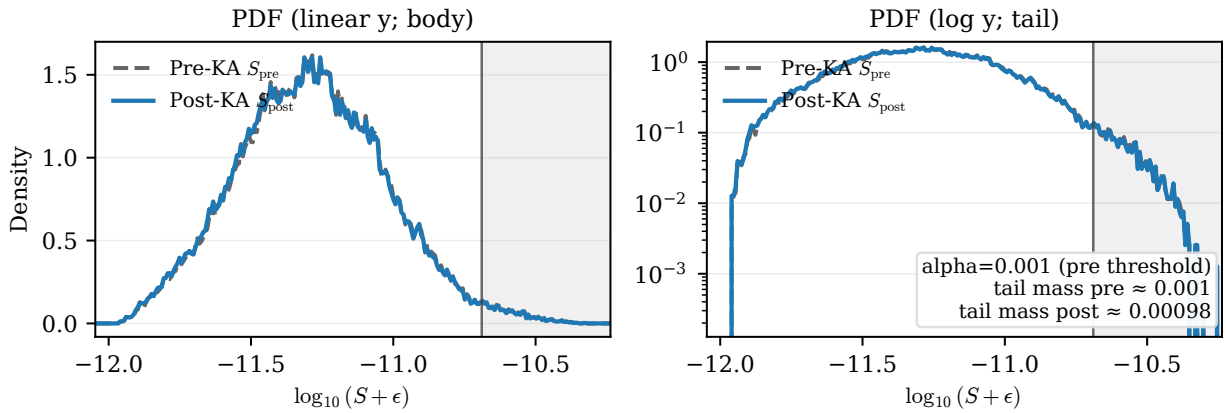

Figure 4: Empirical statistical effect of the strictly bounded, shrink-only penalty on a heavy-tailed background-proxy score distribution (Twinkling calculi artifact sequence). The pre-KA matched-subspace score $S_{\text{stap,pre}}$ exhibits a heavy right tail characteristic of structured artifacts. Applying the non-increasing penalty $\gamma(i) \geq 1$ yields the post-regularization score $S_{\text{stap,post}} = S_{\text{stap,pre}}/\gamma(i)$, compressing the background tail while preserving an explicit do-not-penalize protected set (scores unchanged on high-confidence flow locations). The vertical line indicates the pre-KA threshold τ_α chosen so that $\mathbb{P}\{S_{\text{stap,pre}} \geq \tau_\alpha \mid \text{bg proxy}\} = \alpha$ (here $\alpha = 10^{-3}$); the shaded region highlights the corresponding right tail, and the post-KA tail mass is evaluated at the same τ_α.

4 Experimental Design

4.1 Fixed-profile protocol and operating-point reporting

All hyperparameters (tile geometry, aperture, band definitions, conditioning targets, and prior thresholds) are fixed per regime before reporting results. Detection is evaluated by thresholding the high-score tail (right tail) at fixed FPR targets:

$$\mathbb{P}\{S \geq \tau_\alpha \mid H_0\} = \alpha, \qquad \mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}.$$

For each experiment, we report the supported floor $fpr_{\min} = 1/n_{\mathrm{neg}}$ and bootstrap uncertainty at the natural unit of variation (windows or frames, depending on regime). These choices enforce a strict low-FPR operating discipline and avoid over-interpreting unsupported tail points.

Threshold calibration and deployment. Unless otherwise stated, τ_α is calibrated per window (or per cine sequence) on that unit's negative set to report $\mathrm{TPR}(\alpha)$ at the intended operating point; in labeled regimes, the negative set is available for *exact* calibration, while in deployment it would be estimated via a CFAR-style training procedure or from a fixed calibration set. This per-window protocol should therefore not be read as requiring oracle per-window tuning in deployment. We accordingly report a cross-window threshold-transfer audit for Brain-* pilots (Table 4), which calibrates τ_α on one window and evaluates on a disjoint window and quantifies the resulting realized FPR drift.

4.2 Integrity checks and deterministic sentinels

This work is a reproducible processing pipeline (decode $\rightarrow$ baseline $\rightarrow$ STAP score $\rightarrow$ optional KA regularization). Several practical pathologies can invalidate tail-rate reporting or spuriously activate the KA layer (e.g. missing acquisition metadata, coarse short-ensemble band discretization, proxy-mask circularity, or compute-gating leakage). We therefore enforce deterministic, label-free integrity checks that either (i) record provenance and proceed conservatively or (ii) disable KA ($w \equiv 1$) with an explicit reason label. A concise checklist (and where each sentinel is logged) is provided in Supplementary Material.

4.3 Regimes and claim scope

Claims are stratified by regime to separate quantitative detection efficacy from label-free instrumentation.

- **Brain-k-Wave simulations (labeled).** Across Brain-OpenSkull / Brain-AliasContract / Brain-SkullOR regimes, the fixed profile retains measurable sensitivity at ultra-low FPR in the matched-subspace STAP score (i.e., $\mathrm{TPR}(\alpha) > 0$ as $\alpha \rightarrow 10^{-4}$–$10^{-3}$), with the strongest low-FPR gains in the alias-augmented Brain-AliasContract and skull/OR-inspired Brain-SkullOR regimes (Table 6).
- **Phantom acquisitions (structurally labeled Gammex flow phantom).** We decode raw beamformer outputs, derive B-mode-only tube masks (flow/background), benchmark conventional short-ensemble Doppler baselines, and report structural-label low-FPR operating points and within-ensemble motion ladders.

Regime	Data type	Claim scope
Brain-* (k-Wave)	simulated IQ	low-FPR ROC and latency (simulation)
Gammex flow phantom	real Doppler IQ (phantoms)	structural-label ROC + motion ladders (phantom; not clinical)
ULM Zenodo 7883227	real kHz IQ	motion robustness proxies (no task ROC)
Supplementary datasets	real IQ and PD-only	descriptive/audit analyses (no task ROC)

Table 2: Claims discipline by regime. Quantitative detection performance is reported on labeled Brain-* simulations and on structurally labeled phantom data (Gammex flow phantom). ULM 7883227 is used for label-free motion-robustness evidence on real kHz IQ. Additional datasets are used only for descriptive/audit analyses and are reported in Supplementary Material.

Regime (sampling)	Baseline	STAP core (fixed profile)	Doppler bands (Hz)
Brain-* (k-Wave; PRF 1500; 64-frame windows)	MC–SVD (energy fraction $e = 0.90$); adaptive SVD (SV similarity cutoff); block-wise local SVD (overlap-add; $e = 0.90$); RPCA; HOSVD (baselines scored by PD on residual)	tiles 8×8, stride 3; $L_t = 8$; robust covariance (Tyler-type) with diagonal loading $\lambda = 0.07$; matched-subspace flow-band score	$P_f = [30, 250]$, guard $[250, 400]$, $P_a = [400, 750]$
Gammex flow phantom (linear17; PRF 2500; $N = 17$ shots; pooled over cine frames)	baseline SVD band-pass keep $[2, 17]$; conventional short-ensemble comparators (PD/log-PD/Kasai)	tiles 8×8; $L_t = 16$; Tyler-type covariance with $\lambda = 0.07$; stride 6 (along) / 4 (across); matched-subspace flow-band score	$P_f = [150, 450]$, guard $[450, 700]$, $P_a = [700, 1200]$
ULM Zenodo 7883227 (PRF 1000; 128-frame blocks)	MC–SVD baseline with energy fraction $e = 0.975$ frozen by a one-time label-free sweep	tiles 8×8, stride 3; $L_t = 64$; SCM covariance with $\lambda = 0.07$; matched-subspace flow-band score	$P_f = [10, 150]$, guard $[150, 200]$, $P_a = [200, 500]$

Table 3: Frozen operating profiles for the main regimes. All settings are fixed per regime prior to reporting results; exact commands and exported metadata are provided in Supplementary Material.

- **ULM Zenodo 7883227 (real kHz IQ).** We report label-free motion-robustness curves on beamformed IQ blocks using controlled rigid/nonrigid motion injection, with fairness alignment and degeneracy checks (no task ROC).
- **Supplementary analyses.** Additional label-free audits (e.g. Shin Fig3 real IQ; PD-only whole-brain fUS) and diagnostic panels are provided in Supplementary Material.

5 Results

Results roadmap. We first report labeled Brain-* simulation results at ultra-low FPR, then structurally labeled Gammex flow phantom results (including motion ladders), and finally label-free motion-robustness evidence on real kHz IQ (ULM 7883227). Additional dataset audits and diagnostic panels are provided in Supplementary Material.

Scope and low-FPR operating protocol. Quantitative detection performance is reported on labeled Brain-* simulations and on structurally labeled phantom regimes (Gammex flow phantom sequences; labels derived from B-mode geometry) using the strict low-FPR operating protocol of

Section 4.1 at ultra-low FPR. Real kHz IQ datasets are used for instrumentation (band-energy integrity checks and label-free motion-robustness proxies), not task ROC. Conditional execution is a decision-rule variant (compute gating) that uses a flow-proxy mask; leakage ablations and additional diagnostics are reported in Supplementary Material.

5.1 Brain-k-Wave regimes (labeled): low-FPR ROC and latency

We report labeled low-FPR operating points on Brain-* k-Wave simulations using flow-evidence detection scores under a right-tail convention (higher = more flow-like; thresholds calibrated on the high-score tail): a baseline power Doppler score and the proposed matched-subspace STAP score (Notation). Baseline methods compare clutter suppression strategies (MC–SVD; adaptive SVD with a spatial singular-vector similarity cutoff; block-wise local SVD with overlap-add; RPCA; HOSVD) and are scored using the same power Doppler statistic; STAP is evaluated on top of the MC–SVD residual. Across Brain-* regimes, these baselines (including adaptive and local SVD filters) degrade sharply in the ultra-low-FPR tail, while the localized STAP score retains measurable sensitivity at ultra-low FPR (i.e., $\mathrm{TPR}(\alpha) > 0$ at $\alpha = 10^{-4}$–10^{-3}), including in the SkullOR regime where alias deliberately leaks into the positive class. Table 6 summarizes low-FPR operating points for representative pilot realizations over five non-overlapping 64-frame windows (offsets 0/64/128/192/256); each window has $n_{\mathrm{neg}} = 50199$ background pixels ($fpr_{\min} = 1/n_{\mathrm{neg}} \approx 1.99 \times 10^{-5}$), so FPR targets down to 10^{-4} are empirically supported. Numbers are medians (IQR) over windows and are computed by choosing τ per method per window on that window's negative set to match the target FPR, then evaluating TPR on the corresponding positive set. The wider IQR at $\alpha = 10^{-4}$ in Brain-OpenSkull reflects window-to-window nonstationarity and finite tail support under a fixed, local-covariance profile (some windows with nonzero supported-tail sensitivity versus near-floor windows). Per-window thresholds and realized FPRs are provided in Supplementary Material. Importantly, the MC–SVD+PD baseline row and the STAP row share the identical MC–SVD residual cube and differ only in the detector statistic (PD versus matched-subspace), isolating the effect of the detector rather than the clutter-suppression stage. To quantify deployment-relevant threshold transfer, Table 4 calibrates τ_α on one window and evaluates on a disjoint window, reporting both cross-window TPR and the realized cross-window FPR (FPR drift). Full per-pair outputs are written to `reports/brain_crosswindow_calibration.csv`.

Baseline fairness and frozen settings. All baseline hyperparameters are fixed prior to reporting results. For Brain-* pilots, the MC–SVD energy-fraction cutoff is selected by a one-time sweep and then frozen at $e = 0.90$ for all reported windows (sweep report in `reports/brain_mcsvd_energy_sweep_seed1.csv` / `reports/brain_mcsvd_energy_sweep_seed1.json`). The adaptive similarity-cutoff SVD and block-wise local SVD baselines use deterministic, frozen rules; implementation details are provided in Supplementary Material and Supplementary Material. RPCA and HOSVD baselines use standard, reproducible settings documented in Supplementary Material. All baselines are scored using the same right-tail power Doppler statistic so that differences isolate the clutter-suppression stage rather than the scoring rule. Table 5 summarizes the baseline families and what differs between them.

Detector-component ablation (whitening versus band selectivity). To address the concern that the proposed score is "local adaptive whitening + band energy" rebranded, we report two detector ablations on the identical MC–SVD residual cube: (i) *whitened total power* (apply

Regime	α	within-window TPR med (IQR)	cross-window TPR med (IQR)	cross-window realized FPR med (IQR)
Brain-OpenSkull	10^{-4}	0.4637 (0.3137,0.7647)	0.5534 (0.0503,0.8797)	1.59e-04 (0,2.82e-03)
Brain-OpenSkull	3×10^{-4}	0.5790 (0.5713,0.7682)	0.5898 (0.1243,0.9115)	2.49e-04 (0,8.00e-03)
Brain-OpenSkull	10^{-3}	0.7751 (0.7516,0.7874)	0.6765 (0.3211,0.9379)	1.47e-03 (1.49e-05,3.01e-02)
Brain-AliasContract	10^{-4}	0.0588 (0.0323,0.2284)	0.1999 (0.0003,0.8787)	1.29e-04 (0,5.39e-03)
Brain-AliasContract	3×10^{-4}	0.0634 (0.0557,0.3299)	0.2676 (0.0006,0.9243)	1.13e-03 (0,1.69e-02)
Brain-AliasContract	10^{-3}	0.1619 (0.0665,0.4848)	0.3814 (0.0038,0.9367)	2.02e-03 (0,2.72e-02)
Brain-SkullOR	10^{-4}	0.6082 (0.1561,0.6121)	0.6023 (0.0663,0.8322)	1.20e-04 (0,1.71e-03)
Brain-SkullOR	3×10^{-4}	0.7017 (0.1888,0.7424)	0.6728 (0.0812,0.8800)	3.09e-04 (0,3.33e-03)
Brain-SkullOR	10^{-3}	0.8055 (0.1972,0.8166)	0.7887 (0.1349,0.9126)	1.11e-03 (7.47e-05,3.47e-03)

Table 4: Brain-* cross-window threshold calibration audit for the pre-KA STAP score. "Within-window" matches the main reporting protocol (calibrate τ_α on a window's negatives and evaluate on that same window) and is identical to the STAP row of Table 6 (same score, masks, and five 64-frame windows). "Cross-window" calibrates τ_α on one window and evaluates on a disjoint window ($n = 20$ ordered window pairs). Reported values are median (IQR) over windows/pairs, and the realized cross-window FPR quantifies drift under window-to-window distribution shift.

$R_i^{-1/2}$ but do not partition the Doppler spectrum) and (ii) an *unwhitened* matched-subspace ratio (set $R_i = I$ but retain the fixed flow-band subspace). Table 7 shows that whitening alone without band partition does not control the strict tail, while matched-subspace (banded) scoring is the dominant contributor in these Brain-* regimes under the fixed-profile protocol. The unwhitened ratio corresponds to a normalized Doppler-band energy detector; the full STAP score generalizes this statistic to colored interference via tile-local covariance whitening. In these pilot simulations, the strong performance of the identity-covariance control suggests that after coarse baseline filtering the residual can be close to temporally white; we retain whitening in the primary score as the statistically principled colored-interference generalization.

Detector-swap composability audit (matched-subspace scoring on competing residuals). To isolate detector effects from the clutter-suppression stage, we also compute the matched-subspace STAP score on residuals produced by competing clutter filters (RPCA and HOSVD) and compare it to power Doppler scoring on the same residual under the identical window/mask protocol. Table 8 reports this paired-residual audit; under the fixed STAP profile used throughout, strict-tail improvements do not automatically persist when composing STAP with these more aggressive low-rank baselines. We therefore treat STAP as a second-stage detector defined for a coarse baseline residual (MC–SVD in the main pipeline), and use the controlled detector swap in Table 6 (PD versus matched-subspace on the same MC–SVD residual) as the primary isolation of detector effects.

Covariance-training contamination ablation (Brain-*). Because the detector relies on tile-local covariance estimation, a potential concern is "self-training" or training-set contamination: if high-flow snapshots dominate $\widehat{R}_i$, whitening could suppress the very structure being detected or

Method	scope	cutoff / estimator	tuned?	score	specified
MC–SVD	global	fixed energy fraction ($e = 0.90$)	tune-once	PD	Supplementary Material
Adaptive SVD (SV similarity cutoff)	global	rank via spatial singular-vector similarity drop	deterministic	PD	Supplementary Material
Block-wise local SVD (overlap-add)	local	tile-local energy fraction ($e = 0.90$) + overlap-add	deterministic	PD	Supplementary Material
RPCA	global	PCP (IALM) with fixed iterations/tolerance	deterministic	PD (sparse)	Supplementary Material
HOSVD	global	tensor energy fractions (default $0.99, 0.99, 0.99$)	deterministic	PD (residual)	Supplementary Material
STAP (pre-KA)	local	robust covariance → whitening → flow-subspace ratio	fixed profile	matched-subspace	Section 3

Table 5: Summary of baseline comparators and detector statistics used in the Brain-* labeled regimes. Baseline rows share the same PD flow-evidence score on their respective residual cubes; STAP differs by applying tile-local covariance whitening and a matched-subspace (flow-subspace) score on top of the MC–SVD residual.

	Brain-OpenSkull			Brain-AliasContract			Brain-SkullOR		
Method / score	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}
MC–SVD	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
Adaptive SVD (spatial singular-vector similarity cutoff)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
Block-wise local SVD (overlap-add)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
RPCA	0.0000 (0.0000,0.0277)	0.0000 (0.0000,0.5679)	0.0507 (0.0015,0.9081)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
HOSVD	0.0000 (0.0000,0.0000)	0.0015 (0.0000,0.0031)	0.0062 (0.0031,0.0062)	0.0015 (0.0015,0.0015)	0.0015 (0.0015,0.0015)	0.0046 (0.0046,0.0062)	0.0015 (0.0000,0.0015)	0.0015 (0.0015,0.0031)	0.0062 (0.0046,0.0077)
STAP (pre-KA matched-subspace) on MC–SVD residual	0.4637 (0.3137,0.7647)	0.5790 (0.5713,0.7682)	0.7751 (0.7516,0.7874)	0.0588 (0.0323,0.2284)	0.0634 (0.0557,0.3299)	0.1619 (0.0665,0.4848)	0.6082 (0.1561,0.6121)	0.7017 (0.1888,0.7424)	0.8055 (0.1972,0.8166)

Table 6: Brain-* k-Wave low-FPR operating points using flow-evidence detection scores (higher = more likely flow): baseline power Doppler and matched-subspace STAP. Each regime uses five non-overlapping 64-frame windows (offsets 0/64/128/192/256); each window has $n_{\text{neg}} = 50199$ ($fpr_{\min} \approx 1.99 \times 10^{-5}$). At this support, FPR targets 10^{-4}, 3×10^{-4}, and 10^{-3} correspond to ≈ 5, ≈ 15, and ≈ 50 false-positive pixels per window, respectively. Entries report median (IQR) TPR across windows at each target FPR. Baselines compare clutter suppression strategies (MC–SVD; adaptive SVD (SV similarity cutoff); block-wise local SVD (overlap-add); RPCA; HOSVD) and are scored by power Doppler; STAP is applied on top of MC–SVD. Full per-window outputs are provided in Supplementary Material.

yield brittle thresholds. In the main profile we estimate R_i from all Hankelized snapshots within each tile (no oracle exclusion). As a guard-cell-style ablation, we re-estimate the robust covariance after trimming the top $q = 0.05$ highest-energy training snapshots (per tile) and leave the detector statistic unchanged. Table 9 reports the resulting strict-tail operating points; in these Brain-* pilots the trim ablation does not qualitatively change the supported low-FPR behavior, suggesting that the reported strict-tail gains are not an artifact of covariance self-training. We further report a 1D fixed-profile sensitivity sweep over covariance diagonal loading λ, tile geometry, and slow-time aperture L_t on the canonical Brain-OpenSkull pilot (Supplementary Material).

Relaxed operating-point sanity anchor (Brain-*). Several baseline rows in Table 6 take the value $\text{TPR}(\alpha) \approx 0$ at $\alpha \leq 10^{-3}$. To reduce ambiguity around whether these zeros reflect implementation or tuning errors, Table 10 contrasts one strict-tail operating point ($\alpha = 10^{-3}$) with two relaxed operating points ($\alpha \in \{10^{-2}, 10^{-1}\}$) under the same window protocol; $\alpha = 10^{-2}$ is a moderate reference point and $\alpha = 10^{-1}$ is an intentionally relaxed non-degeneracy sanity anchor.

Why PD-after-filter baselines can collapse in the strict tail. A baseline clutter filter can yield nontrivial separation at moderate operating points yet still fail at ultra-low FPR if the

Method	OpenSkull			AliasContract			SkullOR		
	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$
PD on MC–SVD residual	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
Whitened power (no band)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
Unwhitened ratio (no whitening)	0.9612 (0.9596,0.9627)	0.9792 (0.9712,0.9796)	0.9912 (0.9904,0.9915)	0.9454 (0.9435,0.9504)	0.9623 (0.9612,0.9646)	0.9862 (0.9858,0.9862)	0.9477 (0.9466,0.9489)	0.9592 (0.9585,0.9646)	0.9858 (0.9854,0.9869)
STAP (whitened ratio; pre-KA)	0.4637 (0.3137,0.7647)	0.5790 (0.5713,0.7682)	0.7751 (0.7516,0.7874)	0.0588 (0.0323,0.2284)	0.0634 (0.0557,0.3299)	0.1619 (0.0665,0.4848)	0.6082 (0.1561,0.6121)	0.7017 (0.1888,0.7424)	0.8055 (0.1972,0.8166)

Table 7: Detector ablations on Brain-* regimes (MC–SVD residual, fixed profile). All rows share the identical MC–SVD residual cube and differ only in the detector statistic: baseline PD, total power after tile-local whitening (no Doppler band partition), matched-subspace ratio without whitening (R=I), and the full STAP whitened matched-subspace ratio. Numbers are medians (IQR) over five disjoint 64-frame windows (offsets 0/64/128/192/256).

Method	OpenSkull			AliasContract			SkullOR		
	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$
RPCA+PD (paired)	0.0000 (0.0000,0.0277)	0.0000 (0.0000,0.5679)	0.0507 (0.0015,0.9081)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
RPCA+STAP (paired; pre-KA)	0.0000 (0.0000,0.1972)	0.0000 (0.0000,0.3591)	0.0000 (0.0000,0.6025)	0.0000 (0.0000,0.1780)	0.0000 (0.0000,0.2972)	0.0000 (0.0000,0.4387)	0.0000 (0.0000,0.1169)	0.0000 (0.0000,0.1884)	0.0000 (0.0000,0.2930)
HOSVD+PD (paired)	0.0000 (0.0000,0.0000)	0.0015 (0.0000,0.0031)	0.0062 (0.0031,0.0062)	0.0015 (0.0015,0.0015)	0.0015 (0.0015,0.0015)	0.0046 (0.0046,0.0062)	0.0015 (0.0000,0.0015)	0.0015 (0.0015,0.0031)	0.0062 (0.0046,0.0077)
HOSVD+STAP (paired; pre-KA)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0004)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)

Table 8: Detector-swap fairness check on Brain-* regimes: PD vs matched-subspace STAP detector scores computed on the *same* residual produced by a competing clutter filter (paired bundles). Numbers are medians (IQR) over five disjoint 64-frame windows (offsets 0/64/128/192/256).

background score distribution has an extreme right tail: the background-calibrated threshold τ_α (e.g. the 99.9th percentile of H_0 at $\alpha = 10^{-3}$) can lie above essentially all H_1 pixels, yielding $\mathrm{TPR}(\alpha) \approx 0$ by construction. For example, in a representative Brain-OpenSkull window the MC–SVD+PD baseline background quantiles are $q_{0.99} \approx 5.9$ and $q_{0.999} \approx 9.3$ while the flow $q_{0.999} \approx 0.094$, so no flow pixels exceed the $\alpha = 10^{-3}$ threshold; Figure 5 visualizes the corresponding tail geometry ($\tau_{10^{-3}} > \max(H_1)$) for the baseline PD score and contrasts it with the STAP score computed on the same residual. This tail-dominance failure mode motivates both the strict low-FPR operating discipline and detectors that explicitly reshape the tail (here: tile-local covariance whitening and matched-subspace scoring).

Latency and conditional execution. We also report representative runtimes for baseline clutter suppression and STAP under a conservative fixed operating profile (tile geometry, slow-time aperture L_t, and covariance-conditioning targets fixed *a priori*). Conditional execution reduces runtime by skipping the STAP stage on tiles with negligible proxy flow support and reverting to the baseline score there. Because this changes the overall decision rule, it can change operating characteristics; we therefore treat conditional execution as a distinct variant and require explicit leakage controls when reporting it. Unless otherwise stated, conditional execution uses a *disjoint-window* proxy mask for detection reporting (mask derived from one window and applied to a disjoint window) to avoid same-window proxy leakage.

Portable-workstation feasibility (tempered). A common critique of covariance-adaptive STAP is that local covariance conditioning and the associated matrix factorizations/inversions are prohibitively expensive when applied densely across an image grid. Table 11 shows that, with a batched GPU implementation and steady-state reporting, the baseline stage and the STAP stage are each sub-second on the tested regimes, yielding total baseline+STAP wall time below ≈ 1.3 s per 64-frame Brain-* window and below ≈ 1.0 s per 128-frame Shin window on a laptop-class GPU.[1]

[1]Latency is reported from steady-state replay (windows/frames 2..N) and excludes one-time GPU overheads (initialization, compilation, optional CUDA-graph capture) as well as disk I/O. For context, a 64-frame Doppler window at PRF 1500 Hz corresponds to $64/1500 \approx 43$ ms of acquired slow-time data, and a 128-frame window

Method	OpenSkull			AliasContract			SkullOR		
	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$
STAP (default covariance training)	0.4637 (0.3137,0.7647)	0.5790 (0.5713,0.7682)	0.7751 (0.7516,0.7874)	0.0588 (0.0323,0.2284)	0.0634 (0.0557,0.3299)	0.1619 (0.0665,0.4848)	0.6082 (0.1561,0.6121)	0.7017 (0.1888,0.7424)	0.8055 (0.1972,0.8166)
STAP (cov-trim $q = 0.05$)	0.5136 (0.4506,0.7747)	0.6747 (0.6305,0.7778)	0.7851 (0.7835,0.8343)	0.0692 (0.0396,0.2545)	0.0723 (0.0650,0.3579)	0.1865 (0.0746,0.5110)	0.6171 (0.1742,0.6201)	0.7078 (0.2076,0.7451)	0.8032 (0.2176,0.8266)

Table 9: Covariance-training trim ablation on Brain-* regimes. We compare the default STAP covariance training (all Hankel snapshots) to a trimmed variant that excludes the top-q highest-energy snapshots from the tile-local covariance estimator (q=0.05). Entries are within-window TPR medians (IQR) over five disjoint 64-frame windows, reported at strict-tail FPR targets $\alpha \in \{10^{-4}, 3\times10^{-4}, 10^{-3}\}$.

	Brain-OpenSkull			Brain-AliasContract			Brain-SkullOR		
Method / score	TPR@10^{-3}	TPR@10^{-2}	TPR@10^{-1}	TPR@10^{-3}	TPR@10^{-2}	TPR@10^{-1}	TPR@10^{-3}	TPR@10^{-2}	TPR@10^{-1}
MC–SVD	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0008)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0019)
Adaptive SVD (SV similarity cutoff)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.6417)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.7651)
Block-wise local SVD (overlap-add)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.1461 (0.0834,0.1603)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0596 (0.0473,0.1561)
RPCA	0.0507 (0.0015,0.9089)	0.5917 (0.0334,0.9973)	0.6009 (0.0431,0.9973)	0.0000 (0.0000,0.0000)	0.0031 (0.0000,0.5133)	0.8524 (0.0000,0.9900)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.2253)	0.8462 (0.0000,0.9885)
HOSVD	0.0062 (0.0031,0.0062)	0.0742 (0.0554,0.1030)	0.5244 (0.3914,0.8597)	0.0046 (0.0046,0.0062)	0.0511 (0.0492,0.0846)	0.4333 (0.2857,0.6259)	0.0062 (0.0046,0.0077)	0.0534 (0.0354,0.0861)	0.4314 (0.2549,0.6697)
STAP (pre-KA) on MC–SVD	0.7751 (0.7516,0.7874)	0.8847 (0.7897,0.9508)	0.9996 (0.9812,0.9996)	0.1619 (0.0665,0.4848)	0.7159 (0.0761,0.9950)	0.9050 (0.2245,0.9965)	0.8055 (0.1972,0.8166)	0.9900 (0.2092,0.9938)	0.9996 (0.4283,1.0000)

Table 10: Baseline sanity check on labeled Brain-* simulations at a strict-tail operating point ($\alpha = 10^{-3}$) versus relaxed operating points ($\alpha \in \{10^{-2}, 10^{-1}\}$). Entries report median (IQR) TPR over the same five disjoint 64-frame windows used in the main tables, with thresholds calibrated per window on that window's negative set. The relaxed columns are included only to verify that PD-after-filter baselines are not degenerate; strict-tail reporting remains the focus.

These results suggest that tile-local covariance adaptation need not require server-class hardware and may plausibly fit within portable GPU workstations (e.g. clinical-cart-class systems), subject to end-to-end integration constraints and the decision-rule caveats of conditional execution.
A leakage ablation confirms that conditional execution can inflate apparent performance under same-window proxy gating, while leakage-controlled gating yields a small TPR decrease (order 10^{-3}) at the target FPRs; we therefore treat conditional execution as a distinct decision rule rather than a cost-free speedup. Table 12 summarizes the effect on Brain-OpenSkull pilots; full bootstrap CIs and details are provided in Supplementary Material.

at PRF 1000 Hz corresponds to 128 ms. The present Python prototype therefore does not yet achieve real-time throughput at the full acquisition PRF, but it establishes practical feasibility on laptop-class GPUs and motivates further optimization and tighter scanner integration.

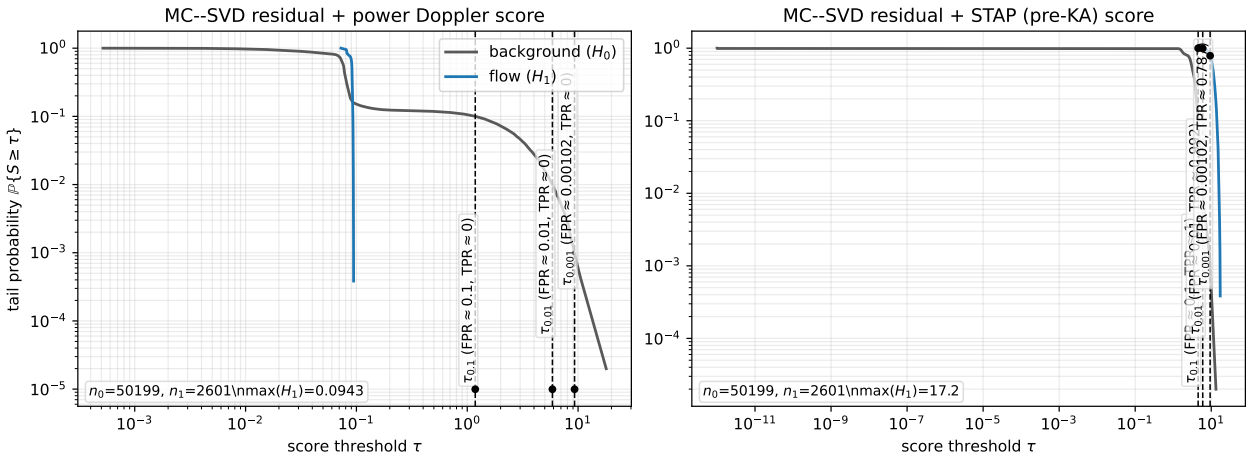

Figure 5: Representative Brain-OpenSkull window illustrating strict-tail collapse of PD-after-filter baselines. Curves show empirical tail probabilities $\mathbb{P}\{S \geq \tau\}$ for background (H_0) and flow (H_1) score distributions versus score threshold τ, for (left) the baseline power Doppler score on the MC–SVD residual and (right) the pre-KA matched-subspace STAP score computed on the same residual. Vertical lines mark background-calibrated thresholds τ_α at $\alpha \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ and markers show the corresponding $\mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}$. For the baseline PD score, $\tau_{10^{-3}}$ lies above $\max(H_1)$, yielding TPR ≈ 0 by construction; the STAP score retains nonzero sensitivity at the same strict-tail operating point.

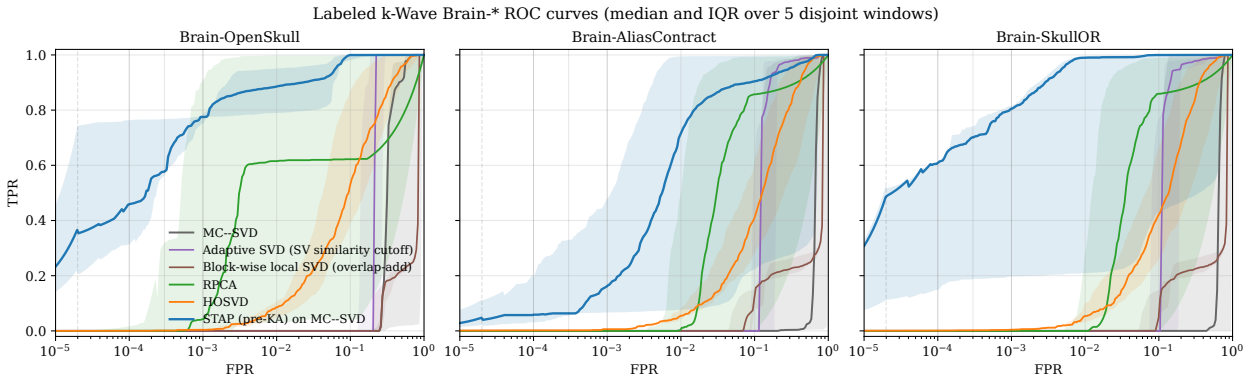

Figure 6: Brain-* k-Wave labeled regimes: ROC-style curves (TPR vs FPR on a log-x axis) under the same 64-frame window protocol used for the Brain-* low-FPR operating-point reporting in the main text. Curves include MC–SVD, adaptive and local SVD baselines, RPCA, HOSVD, and STAP; each curve shows the median and interquartile range over five disjoint windows. The moderate-FPR region illustrates that several PD-after-filter baselines can show separation at relaxed operating points while saturating at strict-tail operating points. The vertical dashed line indicates the supported floor $fpr_{\min} = 1/n_{\mathrm{neg}}$.

Scenario	Baseline (s)	STAP full (s)	Total full (s)	STAP conditional (s)	Total conditional (s)
Brain-OpenSkull (k-Wave; 64-frame window)	0.718	0.513	1.231	0.264	0.982
Brain-AliasContract (k-Wave; 64-frame window)	0.634	0.555	1.189	0.333	0.967
Brain-SkullOR (k-Wave; 64-frame window)	0.617	0.553	1.170	0.314	0.931
Shin RatBrain Fig3 (PRF 1000; 128-frame window; $L_t = 64$)	0.051	0.939	0.990	0.440	0.491
Gammex along (linear17 @ PRF 2500; per cine frame)	0.012	0.246	0.258	—	—
Gammex across (linear17 @ PRF 2500; per cine frame)	0.005	0.105	0.110	—	—

Table 11: Consolidated CUDA-accelerated latency summary (steady-state timing; excludes disk I/O). Times are steady-state means over windows/frames 2..N from a single replay process: Brain-* uses windows 2..5 (offsets 0/64/128/192/256) via `scripts/latency_rerun_check.py`, while Shin and Gammex use windows/frames 2..N via `scripts/latency_realdata_rerun_check.py`. The first replay window/frame includes one-time GPU overheads (initialization, kernel compilation, and optional CUDA-graph capture) and is excluded. "Total" reports baseline+STAP wall time. Hardware: NVIDIA GeForce RTX 3050 6GB Laptop GPU (CUDA 12.1; `stap_device=cuda`). Conditional execution is not reported for structurally labeled Gammex bundles.

Conditional gating mask	ΔTPR@10^{-4} ($\times 10^{-3}$)	ΔTPR@3×10^{-4} ($\times 10^{-3}$)	ΔTPR@10^{-3} ($\times 10^{-3}$)
Same-window proxy mask	$+0.31$	$+0.81$	$+3.05$
Disjoint-window proxy mask	-3.05	-2.90	-1.74
Random mask (matched coverage)	-3.05	-2.97	-2.36

Table 12: Conditional-execution leakage ablation on Brain-OpenSkull pilots (conditional minus full STAP; five independent realizations). Entries report mean ΔTPR at fixed FPR targets. Same-window proxy gating can inflate apparent performance; disjoint-window proxy gating is used as the default leakage-controlled condition for robustness reporting. Full bootstrap CIs and details are provided in Supplementary Material.

5.2 Phantom acquisitions (Gammex flow phantom): structural ROC and within-ensemble motion ladders

Dataset and acquisition parameters. We use the open-access Doppler Twinkling Artifact database [39], which provides complex beamformer output from phantom acquisitions on a clinical scanner. Each cine frame contains a contemporaneous B-mode image and a short-ensemble color-flow (CFM) slow-time block with $N \in \{5, 9, 17\}$ pulses. Because Doppler PRF is not reliably available in the provided metadata for these acquisitions, all Twinkling experiments treat PRF as an explicit experimental assumption frozen per sequence family (listed in Supplementary Material) and enforce band-discretization sanity checks that disable KA ($w \equiv 1$) when discretized Pf/Pa bands are empty or overlapping (Section 3). Decoding/QC details are provided in Supplementary Material (PRF overrides and decoder sanity); reproduction commands and manifests are provided in Supplementary Material.

B-mode-only structural tube masks (independent H_1/H_0). For the Gammex flow phantom sequences we derive evaluation masks from B-mode structure only (no Doppler/STAP outputs). Using acquisition geometry metadata that specifies the CFM region-of-interest (ROI) within the B-mode grid (beam and sample offsets and any beam subsampling), we map the B-mode ROI onto the CFM grid and segment the tube lumen on a median log-envelope reference image. This yields (i) a conservative lumen (flow) mask, eroded to avoid wall pixels and (ii) a background mask that excludes a guard band around the lumen and bright reflectors. Mask provenance and QC metrics are recorded with each experiment; Figure 7 shows representative overlays. These structural labels support a defensible phantom ROC.

Frozen short-ensemble Doppler profile and baseline fairness scores. For the linear-probe Gammex sequences (PRF = 2500 Hz, $N = 17$ shots), we freeze a short-ensemble operating profile: baseline SVD band-pass keep range $[2, 17]$, STAP aperture $L_t = 16$, tiles 8×8, a Tyler-type robust covariance estimator, and diagonal loading $\lambda = 0.07$. To keep the profile auditable while avoiding sample-support limitations on small ROIs and excessive overlap/mixing on large ROIs, we use a deterministic geometry-only stride policy: choose the largest tile stride ≤ 6 such that the implied tile count is ≥ 500. On the two Gammex views this yields stride 6 (along; large ROI) and stride 4 (across; small ROI). Band definitions are frozen to $P_f = [150, 450]$ Hz and $P_a = [700, 1200]$ Hz. For baseline context on short ensembles, we also report conventional comparators derived from the baseline-filtered cube (power Doppler, log power Doppler, and Kasai lag-1 power). Reproducibility details are provided in Supplementary Material.

Structural-label low-FPR ROC (Gammex). We pool pixel scores across cine frames and report low-FPR operating points using right-tail thresholding on pooled background pixels, then evaluate TPR on pooled lumen pixels. In these short-ensemble Doppler acquisitions, the background set includes strong twinkling/artifact pixels whose right-tail scores dominate conventional baselines; consequently, low-FPR thresholds at $\alpha \leq 10^{-3}$ can lie above essentially all lumen pixels, yielding numerically zero baseline TPR at the reported operating points. Figure 8 visualizes this effect for a representative baseline comparator. To avoid treating pooled pixels as independent samples, we report 95% confidence intervals via a frame-level bootstrap: we resample cine frames as the unit of variation while holding the pooled threshold fixed. Table 13 summarizes the resulting structural-label operating points for the two Gammex views under their frozen profiles; full pooled

	Along (linear17 @ PRF 2500)			Across (linear17 @ PRF 2500)		
Method / score	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}
Baseline (power Doppler)	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]
Baseline (log power Doppler)	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]
Baseline (Kasai lag-1 power)	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]
STAP (pre-KA matched-subspace)	0.634 [0.628,0.640]	0.803 [0.798,0.808]	0.946 [0.943,0.948]	0.846 [0.842,0.850]	0.891 [0.888,0.893]	0.938 [0.936,0.940]

Table 13: Gammex flow phantom structural-label operating points using B-mode-derived structural masks. All scores are right-tail (higher indicates more flow evidence). Thresholds are chosen per method on pooled background pixels to match each target FPR and evaluated on pooled lumen pixels. Brackets show 95% frame-bootstrap CIs ($n = 2000$ resamples over cine frames) at the fixed pooled threshold. Pooled background counts are $n_{\mathrm{bg}} = 3.85 \times 10^6$ for the along view ($fpr_{\min} \approx 2.6\times10^{-7}$) and $n_{\mathrm{bg}} = 1.97\times10^6$ for the across view ($fpr_{\min} \approx 5.1\times10^{-7}$). Reproducibility details are provided in Supplementary Material.

summaries (including baseline-score variants and realized thresholds) are provided in Supplementary Material. Because STAP is evaluated on the same baseline-filtered residual cube, the strong STAP TPR at strict FPR rules out the interpretation that the baseline filter simply removed lumen flow; rather, the failure mode is heavy-tailed background scores dominating the right-tail threshold (see Figure 8).

Within-ensemble motion ladders (labeled, real acquisition). To obtain a labeled motion-robustness curve on a real acquisition without external motion ground truth, we inject synthetic *within-ensemble* motion directly into the CFM cube: for each cine frame we apply a per-shot rigid translation $(\Delta y_t, \Delta x_t)$ to the N CFM shots prior to baseline and STAP, with motion kinds `rw` (random walk) and `step`. Because the structural tube masks are derived from B-mode and fixed per frame, this produces a labeled "motion ladder" where the only change is within-ensemble corruption. We generate motion-corrupted ensembles (85 cine frames; amplitudes 0–2 px) and aggregate using frame-level bootstraps for CIs together with an explicit amp=0 equivalence check against the non-motion reference. Figure 11 shows the resulting curves; reproducibility details are provided in Supplementary Material.

Supplementary KA hygiene (artifact-heavy calculi). Additional KA hygiene analyses on artifact-heavy calculi sequences (region/reason summaries and representative pre/post tail examples) are provided in Supplementary Material.

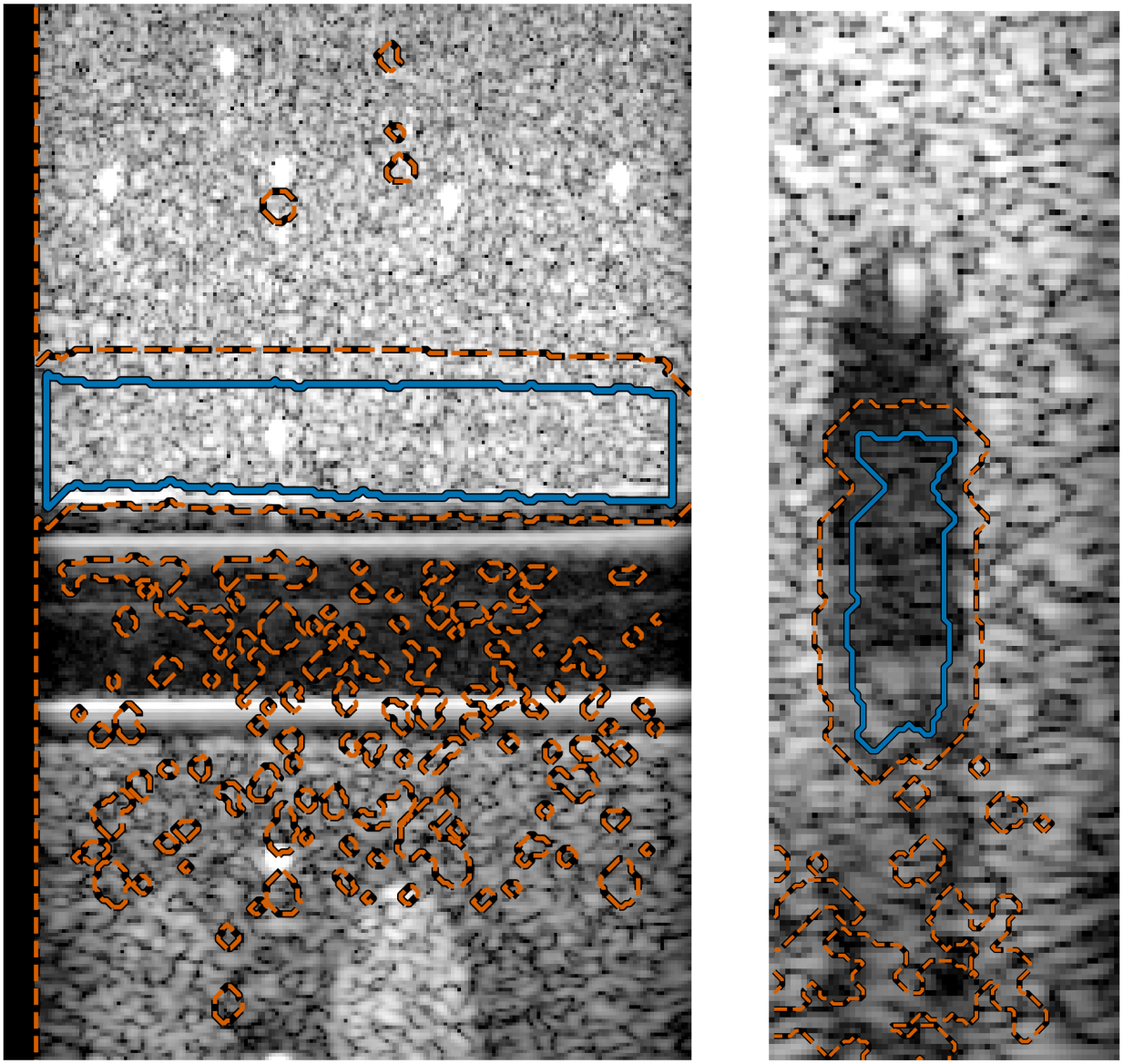

Figure 7: Gammex flow phantom structural evaluation masks derived from B-mode only. Left: along-linear view; right: across-linear view. The lumen mask is intentionally eroded to avoid wall/partial-volume pixels, and the background mask excludes a guard band around the lumen and bright reflectors. Images show the CFM-aligned B-mode ROI with contour overlays of the tube lumen (flow) mask (blue) and background mask (orange, dashed) used for structural-label ROC reporting. Reproducibility details are provided in Supplementary Material.

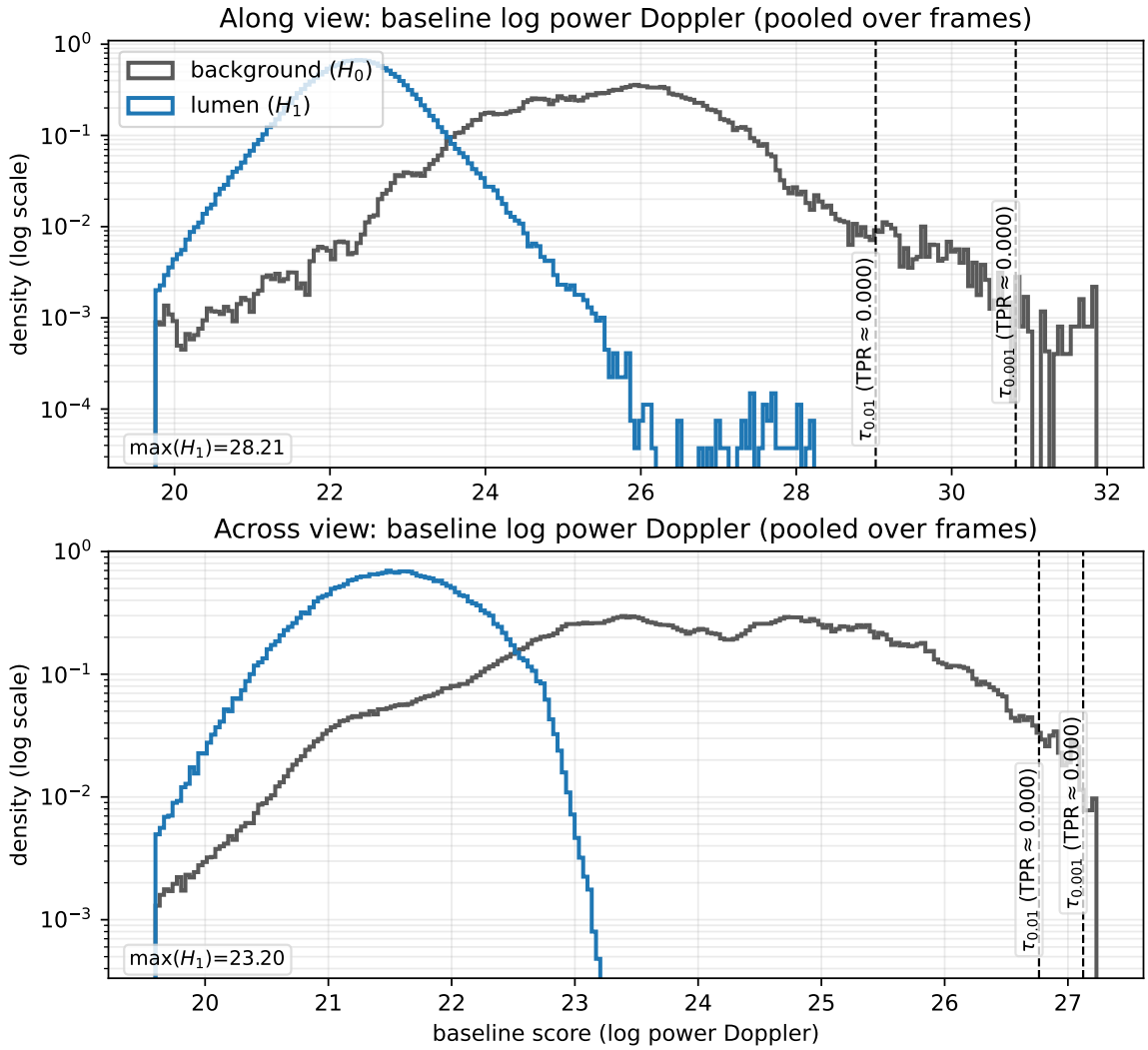

Figure 8: Why conventional baselines can have zero TPR at ultra-low FPR on Gammex short-ensemble data. The plot shows pooled score distributions for the baseline log power Doppler comparator on lumen (H_1; structural tube mask) versus background (H_0; structural background mask), for both along and across views. Vertical lines indicate background-calibrated right-tail thresholds τ_α at $\alpha \in \{10^{-2}, 10^{-3}\}$ (the main text reports $\alpha \leq 10^{-3}$; $\alpha = 10^{-2}$ is shown here for context). On both views the background tail threshold lies above $\max(H_1)$, so $\mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}$ is numerically zero at these tail operating points.

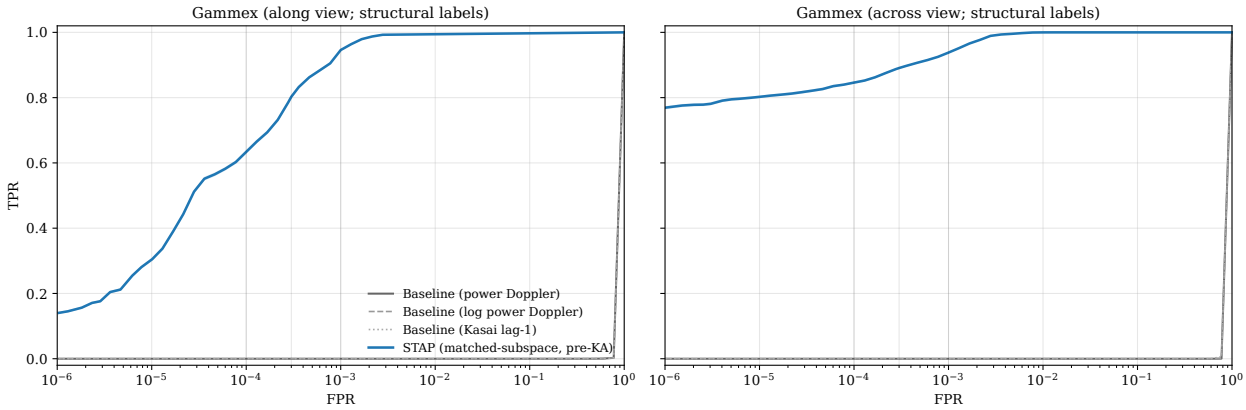

Figure 9: Gammex flow phantom structural labels: pooled ROC-style curves on along and across views, shown for baseline comparators and the pre-KA STAP matched-subspace score. The vertical guides indicate the strict low-FPR operating points reported for the structural-label evaluation in the main text.

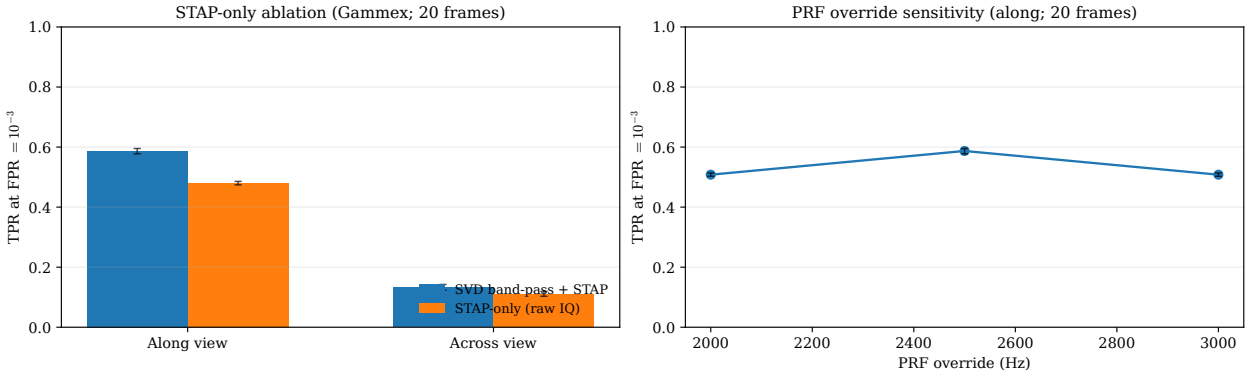

Figure 10: Gammex audits: STAP-only ablation and PRF override sensitivity (first 20 frames; along and across views). Left: STAP-only (raw IQ) vs baseline SVD band-pass + STAP at fixed FPR (10^{-3}). Right: PRF override sensitivity on the along view (PRF $\in \{2000, 2500, 3000\}$ Hz) under a frozen operating profile; this is a low-cost sanity audit of Doppler-band discretization, since PRF is constant within a sequence family and the discretization sentinels depend only on (N, PRF). Additional details are provided in Supplementary Material.

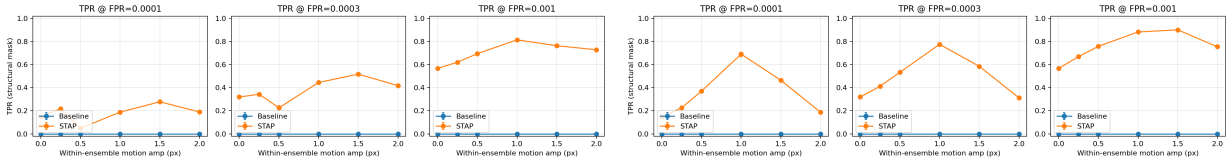

Figure 11: Gammex flow phantom within-ensemble motion ladder on the along-linear PRF=2500 sequence (85 frames). Left: random-walk per-shot translation; right: step translation. Each panel reports TPR vs motion amplitude at fixed FPR targets using B-mode structural masks. Shaded bands are 25–75% bootstrap intervals over cine frames. Reproducibility details are provided in Supplementary Material.

5.3 ULM Zenodo 7883227 (real kHz IQ): baseline calibration and nonrigid motion robustness

Dataset and loader. We use a real kHz IQ dataset released for ultrasound localization microscopy benchmarking [35, 36] and distributed as HDF5 blocks with accompanying acquisition metadata. Each block contains a dataset `iq` of shape $(2, H, W, T) = (2, 256, 180, 800)$ float32 representing I/Q over $T = 800$ slow-time frames. We load blocks on demand and return complex IQ as a cube of shape (T, H, W) (reproduction details in Supplementary Material).

One-time label-free baseline selection. For ULM we select a single MC–SVD energy-fraction cutoff by a one-time label-free sweep on blocks 1–3 over frames 0:128, and then freeze it for all subsequent ULM motion experiments. The sweep uses baseline-only runs and relies on label-free integrity and stability criteria (no task labels); it yields a frozen energy fraction $e = 0.975$ (full sweep details in Supplementary Material).

Nonrigid motion injection and fairness alignment ablation. We evaluate motion robustness by injecting controlled motion directly into the beamformed IQ cube and re-running baseline+STAP under a frozen ULM profile (PRF = 1000 Hz; $L_t = 64$; tiles 8×8; stride 3; SCM covariance; diagonal loading $\lambda = 0.07$; baseline MC–SVD with frozen $e = 0.975$). Because motion is injected at the beamformed IQ stage, this test stresses post-beamforming robustness to residual motion mismatch; it does not model channel-domain decorrelation or skull-induced phase-aberration physics. Motion ladders span two nonrigid families: `brainlike` (rigid random-walk translation plus a smooth, depth-decaying elastic warp with temporally correlated coefficients and micro-jitter) and `elastic` (nonrigid-only; rigid and micro-jitter disabled). For each amplitude we compute label-free robustness endpoints on the baseline score S_{base} and the matched-subspace STAP score $S_{\text{stap,pre}}$ relative to the no-motion reference under three *common* alignment modes (none / common B-mode / injected GT proxy) plus a diagnostic per-method "self" alignment. We additionally report degeneracy checks (std ratios, top-k overlap, and tail quantile ratios) to rule out "stability by collapse." Figure 12 shows the main `brainlike` sweep; additional motion variants are provided in Supplementary Material.

Endpoint definition. For a score map $S_{\text{meth}}(a)$ at motion amplitude a and the corresponding no-motion reference $S_{\text{meth}}(0)$ (same block), we compute $\rho_{\text{meth}}(a)$ as the Pearson correlation between the aligned maps on a central crop Ω_a that excludes a margin of $\lceil a \rceil + 2$ pixels to avoid boundary artifacts from warping. Correlations are computed directly on the exported right-tail scores (no log transform) and use the same crop/alignment for baseline and STAP in the "common" alignment modes.

Representative quantitative endpoints (ULM). At a 3 px motion amplitude under the injected GT alignment proxy, the median correlation on the baseline score S_{base} drops to ≈ 0.20 (`brainlike`), while the STAP score $S_{\text{stap,pre}}$ remains ≈ 0.75 (medians over blocks 1–3; full tabular summaries and additional motion variants are provided in Supplementary Material). Degeneracy checks confirm that this stability is not explained by map collapse: the median std ratio and 0.999-quantile ratio of $S_{\text{stap,pre}}$ remain nontrivial across amplitudes, and top-k overlap decays gradually rather than saturating to trivial values.

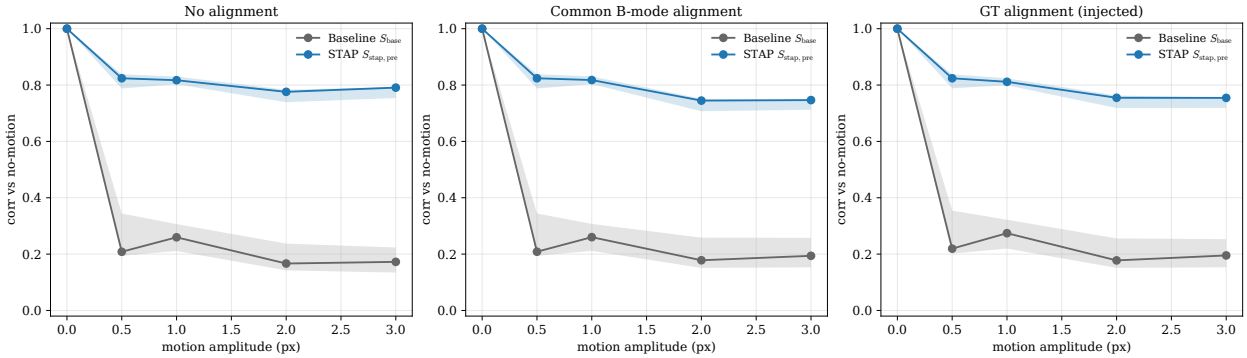

Figure 12: ULM 7883227 motion sweep (blocks 1–3; frames 0:128; frozen baseline $e = 0.975$): `brainlike` motion (rigid + elastic). Each panel reports correlation vs the no-motion reference under a different alignment mode (none / common B-mode / injected GT proxy) for the baseline score S_{base} and the STAP score $S_{\text{stap,pre}}$; shaded bands show median±IQR over blocks. Reproducibility details are provided in Supplementary Material.

Supplementary datasets and diagnostics. Additional label-free audits (e.g. Shin Fig3 real IQ; PD-only whole-brain fUS) and diagnostic panels (e.g. conditional-execution leakage ablations, KA hygiene panels) are provided in Supplementary Material.

6 Discussion

A) Robustness in the ultra-low-FPR tail

This work adapts space–time adaptive processing (STAP), a foundational approach in radar target detection under heavy clutter, to functional ultrasound (fUS) slow-time flow detection. Under the strict low-FPR operating protocol (Section 4.1), we find that common global or semi-global clutter suppressors (MC–SVD, RPCA, HOSVD) can lose sensitivity in the extreme right tail, while the proposed tile-based matched-subspace STAP score retains measurable sensitivity at ultra-low FPR (i.e., nonzero TPR(α) at $\alpha \approx 10^{-4}$–10^{-3} under empirically supported tail floors). This behavior is demonstrated on labeled k-Wave simulations and structurally labeled *in vitro* phantoms, where flow/background masks are defined independently from Doppler/STAP outputs via simulator truth or B-mode geometry. The within-ensemble motion ladders on phantom acquisitions further indicate that STAP retains detection stability under controlled per-shot motion corruption, consistent with its reliance on localized covariance rather than global low-rank structure.

B) Knowledge-aided prior as a fail-safe

The optional knowledge-aided (KA) layer is designed as a conservative safety regularizer rather than an unconstrained signal amplifier. It acts as a shrink-only prior penalty with an auditable (region, reason) output: when spatial-spectral evidence is flat or non-actionable, the prior defaults to $\mathcal{R}_0$ (uninformative; $w \equiv 1$) and leaves the pre-KA detector score unchanged. When the prior lies in $\mathcal{R}_1/\mathcal{R}_2$, it preserves an explicit do-not-penalize protected set (scores unchanged on high-confidence flow locations) and applies only non-increasing shrink on high-risk regions, so that "KA does nothing" is an intended success mode out of regime. This posture prioritizes transparent regime diagnostics and avoids score inflation in regimes where spectral evidence cannot justify aggressive intervention.

C) Data limitations: missing raw high-PRF complex IQ

A key limitation in evaluating Doppler-side covariance detectors is field-wide data availability: most public *in vivo* fUS releases do not include raw, high-PRF complex IQ ensembles needed to evaluate covariance-based Doppler detectors with independent H_1/H_0 labels. Many releases are temporally downsampled, post-filtered, or provided only as power-Doppler equivalents, which can irreversibly destroy the high-rate complex phase history needed for spatiotemporal covariance estimation and band-energy integrity checks. Accordingly, while we report real-data instrumentation (band-energy integrity, prior-regime behavior, and label-free stability under controlled motion corruption), comprehensive *in vivo* ROC evaluation of Doppler-side STAP/KA requires high-PRF raw IQ acquisitions prior to proprietary clutter suppression, with reliable metadata and task/landmark structure to define independent evaluation labels.

7 Conclusion and future work

This study establishes a deterministic, auditable KA-STAP framework for fUS slow-time flow detection under a tune-once-then-freeze discipline. Using a strict low-FPR operating protocol and independent structural labels where available, we show that localized matched-subspace STAP can

preserve separability in regimes where conventional low-rank baselines degrade in the extreme tail, while the KA spatial-spectral prior provides shrink-only, protected-set-preserving risk control with explicit out-of-regime behavior.

Future work is to (i) evaluate Doppler-side STAP/KA on pre-registered, labeled raw-IQ *in vivo* datasets that preserve raw, high-PRF complex IQ phase history and support independent task/landmark labels, (ii) reduce latency via hardware-aware implementations of the batched covariance and band-energy kernels toward real-time deployment, and (iii) explore compact, frozen learned regularizers only when they can be constrained to preserve the shrink-only auditability and low-FPR reporting guarantees established here.

Data and code availability

All datasets used in this work are either publicly available (as cited) or derived deterministically from those sources. All code used to generate the reported tables and figures is versioned; the exact commit hash (and, when available, a public repository URL and release tag) are recorded in the reproducibility manifest (Supplementary Material). A minimal CUDA-enabled software environment is specified in `environment.yml` (Python 3.11; PyTorch 2.4 + CUDA 12.1); a container recipe is provided as `Dockerfile`. As a single-command example, Table 6 can be regenerated (pilot generation + baseline matrix + LaTeX table) by running `bashscripts/reproduce_table5_brain_kwave.sh` (expected output: `reports/brain_kwave_vnext_baselines_table.tex`).

References

[1] Imbault, M., et al. Intraoperative functional ultrasound imaging of human brain activity. *Scientific Reports* 7, 7304 (2017).

[2] Macé, E., Montaldo, G., Pernot, M., Cohen, I., and Tanter, M. Functional ultrasound imaging of the brain. *NeuroImage* 54(4), 2720–2731 (2011).

[3] Urban, A., et al. In vivo whole brain microvascular imaging in mice using ultrafast ultrasound. *EBioMedicine* 76, 103861 (2022).

[4] Alexandrov, A. V., et al. Velocity measurements in the middle cerebral arteries of healthy adults by transcranial Doppler. *Journal of Neuroimaging* 1, 166–172 (1991).

[5] Alexandrov, A. V., et al. Optimal values of flow velocity on transcranial Doppler in patients with intracranial arterial stenosis. *Stroke* 33, 1757–1761 (2002).

[6] Macé, E., et al. Functional ultrasound imaging of the brain: theory and basic principles. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 60(3), 492–506 (2013).

[7] Rabut, C., et al. Functional ultrasound imaging of human brain activity through the skull. *Science Translational Medicine* 12(572), eaav4392 (2020).

[8] Rabut, C., Norman, S. L., Griggs, W. S., Russin, J. J., Jann, K., et al. Functional ultrasound imaging of human brain activity through an acoustically transparent cranial window. *Science Translational Medicine* 16, eadj3143 (2024).

[9] Deffieux, T., Demené, C., and Tanter, M. Functional ultrasound imaging: A new imaging modality for neuroscience. *Neuroscience* 474, 110–121 (2021).

[10] Demeulenaere, O., Bertolo, A., Pezet, S., Osmanski, B.-F., Tanter, M., and Deffieux, T. In vivo whole brain microvascular imaging in mice using transcranial 3D ultrasound localization microscopy. *EBioMedicine* 79, 103995 (2022).

[11] Demené, C., et al. Spatiotemporal clutter filtering of ultrafast ultrasound data highly increases Doppler and fUltrasound sensitivity. *IEEE Transactions on Medical Imaging* (2015). DOI: `https://doi.org/10.1109/TMI.2015.2428634`.

[12] Baranger, J., et al. Adaptive spatiotemporal SVD clutter filtering for ultrafast Doppler imaging using similarity of spatial singular vectors. *IEEE Transactions on Medical Imaging* (2018). DOI: `https://doi.org/10.1109/TMI.2018.2789499`.

[13] Yu, A. C. H., and Løvstakken, L. Eigen-based clutter filter design for ultrasound color flow imaging: a review. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2010). DOI: `https://doi.org/10.1109/TUFFC.2010.1521`.

[14] Song, P., et al. Ultrasound small vessel imaging with block-wise adaptive local clutter filtering. *IEEE Transactions on Medical Imaging* (2017). DOI: `https://doi.org/10.1109/TMI.2016.2605819`.

[15] Chee, A. J. Y., Yiu, B. Y. S., and Yu, A. C. H. A GPU-parallelized eigen-based clutter filter framework for ultrasound color flow imaging. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2017). DOI: `https://doi.org/10.1109/TUFFC.2016.2606598`.

[16] Lok, U.-W., Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Real time SVD-based clutter filtering using randomized SVD and spatial downsampling for microvessel imaging. *Ultrasonics* 107, 106163 (2020). DOI: `https://doi.org/10.1016/j.ultras.2020.106163`.

[17] Demene, C., et al. Improved color Doppler for cerebral blood flow axial velocity estimation (iCD-fUS). *IEEE Transactions on Medical Imaging* 40(5), 1498–1510 (2021).

[18] Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Improved color Doppler for cerebral blood flow axial velocity imaging. *IEEE Transactions on Medical Imaging* 40(2), 758–764 (2021).

[19] Urban, A., et al. Mobile human brain imaging using functional ultrasound. *Nature Biomedical Engineering* 8, 534–546 (2024).

[20] Aaslid, R., Markwalder, T.-M., and Nornes, H. Noninvasive transcranial Doppler ultrasound recording of flow velocity in basal cerebral arteries. *Journal of Neurosurgery* 57(6), 769–774 (1982).

[21] Gao, S., Lam, W. W.-K., Chan, Y.-H., Liu, J., Wong, K.-P., Leung, T. W.-S., et al. Optimal values of flow velocity on transcranial Doppler in grading middle cerebral artery stenosis in comparison with magnetic resonance angiography. *Journal of Neuroimaging* 12(3), 213–218 (2002).

[22] Estrada, H., Gottschalk, S., Reiss, M., Neuschmelting, V., Razansky, D., et al. Observation of guided acoustic waves in a human skull. *Ultrasound in Medicine & Biology* 44(7), 1438–1447 (2018).

[23] Candès, E. J., Li, X., Ma, Y., and Wright, J. Robust principal component analysis? *Journal of the ACM* 58(3), 11 (2011).

[24] Solomon, O., et al. Deep unfolded robust PCA with application to clutter suppression in ultrasound. *IEEE Transactions on Medical Imaging* 39(4), 1051–1063 (2020). DOI: `https://doi.org/10.1109/TMI.2019.2941271`.

[25] Lin, Z., Chen, M., and Ma, Y. The augmented Lagrange multiplier method for exact recovery of corrupted low-rank matrices. *arXiv preprint* arXiv:1009.5055 (2010).

[26] Halko, N., Martinsson, P. G., and Tropp, J. A. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review* 53(2), 217–288 (2011).

[27] Functional ultrasound imaging of the brain. UNC/NCSU Joint Department of Biomedical Engineering teaching material, online resource, accessed 2024. Available at `https://bme.unc.edu/wp-content/uploads/sites/917/2021/08/functional_ultrasound.pdf`.

[28] Soloukey, S., Vincent, A. J. P. E., Smits, M., Slump, C. H., Dirven, C. M. F., et al. Functional ultrasound (fUS) during awake brain surgery: the clinical potential for intra-operative functional and vascular brain mapping. *Frontiers in Neuroscience* 13, 1384 (2019). Available at `https://www.frontiersin.org/articles/10.3389/fnins.2019.01384`.

[29] Rapid full-wave phase aberration correction method for transcranial ultrasound. Online resource, accessed 2024. Available at `https://jtultrasound.biomedcentral.com/articles/10.1186/s40349-016-0074-7`.

[30] Ultrafast Ultrasound Imaging. Online resource, accessed 2024. Available at `https://mdpi-res.com/bookfiles/book/756/Ultrafast_Ultrasound_Imaging.pdf`.

[31] Staggered multiple-PRF ultrafast color Doppler. Online resource, accessed 2024. Available at `https://www.biomecardio.com/publis/ieeetmi16.pdf`.

[32] Functional ultrasound imaging: a useful tool for neuroscience. Online resource, accessed 2024. Available at `https://www.sciencedirect.com/science/article/pii/S1053811921009940`.

[33] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. Zenodo dataset, DOI 10.5281/zenodo.10711806 (2024).

[34] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. *Nature Communications* (2024). Available at `https://www.nature.com/articles/s41467-024-47154-2`.

[35] Chavignon, A., Heiles, B., Hingot, V., Lopez, P., Teston, E., and Couture, O. In vivo rat brain for Ultrasound Localization Microscopy: raw and beamformed data. Zenodo dataset, DOI 10.5281/zenodo.7883227 (2023).

[36] Heiles, B., Chavignon, A., Hingot, V., Lopez, P., Teston, E., and Couture, O. Performance benchmarking of microbubble-localization algorithms for ultrasound localization microscopy. *Nature Biomedical Engineering* (2021). DOI: `https://doi.org/10.1038/s41551-021-00824-8`.

[37] Errico, C., et al. Ultrafast ultrasound localization microscopy for deep super-resolution vascular imaging. *Nature* 527(7579), 499–502 (2015). DOI: `https://doi.org/10.1038/nature16066`.

[38] Hingot, V., et al. Subwavelength motion-correction for ultrafast ultrasound localization microscopy. *Ultrasonics* 77, 17–21 (2017). DOI: `https://doi.org/10.1016/j.ultras.2017.01.008`.

[39] Leonov, D. V., Reshetnikov, R. V., Kulberg, N. S., Nasibullina, A. A., and Gromov, A. I. Doppler Twinkling Artifact Observations: an Open-Access Database of Raw Ultrasonic Signals. *Digital Diagnostics* (2021). DOI: `https://doi.org/10.17816/DD76511`.

[40] Ozgun, K., and Byram, B. C. Multidimensional clutter filtering of aperture domain data for improved blood flow sensitivity. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 68(8), 2645–2656 (2021). DOI: `https://doi.org/10.1109/TUFFC.2021.3073292`.

[41] Huang, Z., et al. Adaptive higher-order singular value decomposition clutter filter for ultrafast Doppler imaging of coronary flow under non-negligible tissue motion. *Ultrasonics* (2024). DOI: `https://doi.org/10.1016/j.ultras.2024.107307`.

[42] Urban, A., et al. Real-time imaging of brain activity in freely moving rodents using functional ultrasound. *Nature Methods* 12(9), 873–878 (2015). DOI: `https://doi.org/10.1038/nmeth.3482`.

[43] Demené, C., et al. 4D microvascular imaging based on ultrafast Doppler tomography. *NeuroImage* 127, 472–483 (2016). DOI: `https://doi.org/10.1016/j.neuroimage.2015.11.014`.

[44] Slepian, D. Prolate spheroidal wave functions, Fourier analysis, and uncertainty—V: The discrete case. *Bell System Technical Journal* 57(5), 1371–1430 (1978). DOI: `https://doi.org/10.1002/j.1538-7305.1978.tb02104.x`.

[45] Thomson, D. J. Spectrum estimation and harmonic analysis. *Proceedings of the IEEE* 70(9), 1055–1096 (1982). DOI: `https://doi.org/10.1109/PROC.1982.12433`.

[46] Percival, D. B., and Walden, A. T. *Spectral Analysis for Physical Applications: Multitaper and Conventional Univariate Techniques*. Cambridge University Press (1993).

[47] Guerci, J. R. *Space-Time Adaptive Processing for Radar*. Artech House (2003).

[48] Wicks, M. C., Rangaswamy, M., Adve, R., and Hale, T. B. Space-time adaptive processing: a knowledge-based perspective for airborne radar. *IEEE Signal Processing Magazine* 23(1), 51–65 (2006). DOI: https://doi.org/10.1109/MSP.2006.1593337.

[49] Melvin, W. L., and Guerci, J. R. Knowledge-aided signal processing: a new paradigm for radar and other advanced sensors. *IEEE Transactions on Aerospace and Electronic Systems* 42(3), 983–996 (2006). DOI: https://doi.org/10.1109/TAES.2006.248215.

[50] Gurram, P. R., and Goodman, N. A. Spectral-domain covariance estimation with a priori knowledge. *IEEE Transactions on Aerospace and Electronic Systems* 42, 1010–1020 (2006). DOI: https://doi.org/10.1109/TAES.2006.248209.

[51] Van Veen, B. D., and Buckley, K. M. Beamforming: a versatile approach to spatial filtering. *IEEE ASSP Magazine* 5(2), 4–24 (1988). DOI: https://doi.org/10.1109/53.665.

[52] Kuglin, C. D., and Hines, D. C. The phase correlation image alignment method. In *Proceedings of the IEEE International Conference on Cybernetics and Society* (1975).

[53] Stone, H. S., Orchard, M. T., Chang, E.-C., and Martucci, S. A. A fast direct Fourier-based algorithm for subpixel registration of images. *IEEE Transactions on Geoscience and Remote Sensing* 39(10), 2235–2243 (2001). DOI: https://doi.org/10.1109/36.957286.

[54] Foroosh, H., Zerubia, J. B., and Berthod, M. Extension of phase correlation to subpixel registration. *IEEE Transactions on Image Processing* 11(3), 188–200 (2002). DOI: https://doi.org/10.1109/83.988953.

[55] Capon, J. High-resolution frequency-wavenumber spectrum analysis. *Proceedings of the IEEE* 57(8), 1408–1418 (1969).

[56] Frost, O. L. An algorithm for linearly constrained adaptive array processing. *Proceedings of the IEEE* 60(8), 926–935 (1972).

[57] Kay, S. M. *Fundamentals of Statistical Signal Processing, Volume II: Detection Theory.* Prentice Hall (1998).

[58] Scharf, L. L., and Friedlander, B. Matched subspace detectors. *IEEE Transactions on Signal Processing* 42(8), 2146–2157 (1994).

[59] Ward, J. Space-time adaptive processing for airborne radar. MIT Lincoln Laboratory, Tech. Rep. 1015 (1994).